



Communication-Adaptive Gradient Sparsification for Federated Learning with Error Compensation

Item Type	Article
Authors	Vaishnav, Shubham;Khirirat, Sarit;Magnússon, Sindri
Citation	Vaishnav, S., Khirirat, S., & Magnússon, S. (2024). Communication-Adaptive Gradient Sparsification for Federated Learning with Error Compensation. IEEE Internet of Things Journal, 1–1. https://doi.org/10.1109/jiot.2024.3490855
Eprint version	Post-print
DOI	10.1109/jiot.2024.3490855
Publisher	Institute of Electrical and Electronics Engineers (IEEE)
Journal	IEEE Internet of Things Journal
Rights	(c) 2024 IEEE. Personal use of this material is permitted. Permission from IEEE must be obtained for all other users, including reprinting/ republishing this material for advertising or promotional purposes, creating new collective works for resale or redistribution to servers or lists, or reuse of any copyrighted components of this work in other works.
Download date	2025-11-24 09:53:12
Link to Item	http://hdl.handle.net/10754/701158

Communication-Adaptive Gradient Sparsification for Federated Learning with Error Compensation

Shubham Vaishnav, Sarit Khirirat, and Sindri Magnússon

Abstract—Federated learning has emerged as a popular distributed machine-learning paradigm. It involves many rounds of iterative communication between nodes to exchange model parameters. With the increasing complexity of ML tasks, the models can be large, having millions of parameters. Moreover, edge and IoT nodes often have limited energy resources and channel bandwidths. Thus, reducing the communication cost in Federated Learning is a bottleneck problem. This cost could be in terms of energy consumed, delay involved, or amount of data communicated. We propose a communication cost-adaptive model sparsification for Federated Learning with Error Compensation. The central idea is to adapt the sparsification level in run-time by optimizing the ratio between the impact of the communicated model parameters and communication cost. We carry out a detailed convergence analysis to establish the theoretical foundations of the proposed algorithm. We conduct extensive experiments to train both convex and non-convex machine learning models on a standard dataset. We illustrate the efficiency of the proposed algorithm by comparing its performance with three baseline schemes. The performance of the proposed algorithm is validated for two communication models and three cost functions. Simulation results show that the proposed algorithm needs a substantially less amount of communication than the three baseline schemes while achieving the best accuracy and fastest convergence. The results are consistent for all the considered cost models, cost functions, and ML models. Thus, the proposed FL-CATE algorithm can substantially improve the communication efficiency of federated learning, irrespective of the ML tasks, costs, and communication models.

Index Terms—Federated learning, Communication efficiency, IoT, Gradient sparsification, Distributed learning.

I. INTRODUCTION

FEDERATED LEARNING (FL) embodies a transformative shift in machine learning (ML), introducing a decentralized methodology for model training that simultaneously upholds data privacy across diverse client nodes. This paradigm enables the collaborative construction of a unified ML model adept at handling heterogeneous datasets from various clients — a feat unattainable in traditional centralized ML frameworks. In contrast to the conventional approach of aggregating extensive datasets at a central server for model training, FL's decentralized nature inherently supports a reduction in overall communication requirements. Notably, this reduction

is not merely a by-product but a core feature, facilitating efficient data utilization without compromising individual data privacy. However, despite these advancements, communication efficiency remains a critical challenge in FL. FL algorithms require multiple communication rounds, with frequent transmission of large ML models, often consisting of millions of parameters. This extensive and repetitive model transmission, particularly over IoT networks with limited bandwidth or intermittent connectivity, can result in significant network traffic and latency, thereby creating substantial delays and reducing the effectiveness of model training. An example of this issue is seen in the energy dynamics of data transmission: sending a single bit of data over wireless channels consumes energy at least 480 times that required for executing one CPU addition instruction [1]. Therefore, optimizing communication efficiency is not merely advantageous but crucial for the scalable and pragmatic deployment of FL, especially in IoT settings with limited network resources.

Recent research increasingly focuses on methods for compressing models or gradients during training to enhance the communication efficiency of FL algorithms. A prevalent technique is sparsification, which involves transmitting only the most significant portions of the ML model. For instance, after a client node updates its model, it only sends the parts of the model that undergo the most substantial changes during the local training. This selective approach capitalizes on the insight that not all model weights are equally significant; some weights play a more pivotal role and undergo more rapid changes during the training process, as expounded, e.g., in [2]. However, this approach can lead to slower convergence and reduced accuracy due to the potential loss of information in the less emphasized weights. To counterbalance these negative impacts, error compensation mechanisms are suggested. These mechanisms aim to rectify cumulative compression errors by incorporating the memory of prior errors, thus correcting the progressive accumulation of inaccuracies. Two notable variants of these mechanisms are error compensation [3]–[5] and EF21 [6].

Fine-tuning the compression level is essential to maximize the efficacy of compression techniques in FL. Achieving an optimal balance is critical: excessive compression risks performance degradation, whereas inadequate compression leads to unnecessary data transmission. Sparsification necessitates carefully determining the right amount of significant model parameters to be transmitted during each communication round. Traditional methods for adjusting compression levels, as implemented in various sparsification approaches [7]–[9], typically rely on hyperparameter optimization before training.

This work is supported by Digital Futures through the project DEMOCRITUS and the Swedish Research Council (Vetenskapsrådet) under grant 2020-03607.

S. Vaishnav and S. Magnússon are with the Department of Computer and System Science, Stockholm University, 11419 Stockholm, Sweden (e-mail: shubham.vaishnav@dsv.su.se; sindri.magnusson@dsv.su.se).

S. Khirirat is with King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia (e-mail: Khirirat.s@gmail.com).

This pre-training optimization, while useful, can be resource-heavy and may not adequately address the fluid nature of FL environments. The appropriate compression level is typically not static; it can vary based on the specific data of each local training set and the models in use, potentially shifting with each iteration. Moreover, it is crucial to factor in the transmission costs and evolving network characteristics when evaluating communication efficiency. As these network conditions and datasets evolve, static sparsification levels risk becoming inefficient, particularly in environments with fluctuating wireless connectivity. This dynamic landscape highlights the necessity for adaptive sparsification methods in FL. Such adaptive techniques should be capable of continuously recalibrating the sparsification level, aligning it with the prevailing network status and communication overheads.

Contributions. We propose a novel adaptive and communication-aware compression framework FL-CATE that adapts both the models that are being transmitted as well as the communication-related costs. In particular, in each iteration of the FL algorithm, each participating client node automatically tunes its sparsification level. This tuning aims to optimize communication efficiency, defined as the ratio of the impact of the retained model parameters after compression to the associated communication cost. The versatility of the FL-CATE framework is one of its key strengths, offering compatibility with a wide range of communication costs and compression models. In this work, we specifically delve into scenarios where communication costs are characterized by the transmitted bits, energy, and latency metrics. A significant theoretical contribution of this paper is the establishment of convergence proofs for FL-CATE in both convex and non-convex problem settings. Our empirical investigation encompasses a comprehensive array of experiments, incorporating three distinct communication cost types, two ML model architectures (covering both convex and non-convex formulations), and two communication models. In benchmark comparisons, which include scenarios with optimally tuned static compression levels, FL-CATE demonstrates substantial performance enhancements. These findings underscore the efficacy of dynamically adjusting compression levels, highlighting FL-CATE as a significant advancement in the realm of FL.

The most closely related work to this paper is our previous publication [10]. However, this paper introduces major novelties and represents a substantial advancement over [10], as listed below:

- We conduct a comprehensive theoretical analysis of the FL-CATE algorithm, providing robust convergence guarantees. This level of theoretical scrutiny was absent in the conference paper [10].
- We incorporate error compensation mechanisms in our approach to effectively counterbalance the potential negative effects of employing compression techniques.
- We notably expand the scope of our experimental evaluation, all the experiments that appear here are new. While the conference version focused solely on convex optimization models, this journal version encompasses experiments with both convex and non-convex model

training, thereby broadening the applicability and relevance of our findings.

- We conduct a more thorough experimental analysis by considering three distinct metrics of communication costs: transmitted bits, latency, and energy. In contrast, the conference paper limited its exploration to energy costs alone.

These enhancements collectively mark a significant leap in our research, presenting a more robust, comprehensive, and versatile exploration of the FL-CATE framework. Furthermore, this journal version significantly expands and enriches all discussions and descriptions.

II. RELATED WORK

Communication bandwidth poses a significant bottleneck in distributed learning systems due to the extensive transmission of high-dimensional information from numerous compute nodes to a central server. This necessitates the development of approaches to reduce the information communicated between the nodes and the central server. Two commonly used approaches are: (1) **Federated optimization** that is based on introducing increased local data processing on each node to reduce the communication need and (2) **Communication-efficient optimization** that applies compression operators on the communicated information to reduce the transmission load per communication round. In this section, we review previous literature on communication-efficient optimization methods and federated optimization methods that are closely related to our work.

Federated optimization: Federated optimization methods reduce communication by increasing local computation in each node before it transmits the information to the central server. Popular federated optimization methods include FedAvg (or Local SGD) [11] and FedProx [12]. FedProx is a generalization of FedAvg, introduced to tackle the heterogeneity in federated networks [12]. These methods have been extensively studied in various settings and modified to improve their convergence behaviors in theory and practice. Some of the approaches used to modify these preliminary approaches are discussed here. One approach is to reorganize the methods using operator-splitting schemes (e.g., FedSplit) [13], ADMM formulation (e.g., FedPD [14]), and ProxSkip (called Scaffnew) [15]. FedPAQ [16] benefits from using compression operators and local updates to save communications. FedPAQ has been modified to improve the convergence using error compensation, e.g., Q-Sparse LocalSGD [17], and FedGate [18].

Communication-efficient optimization: Sparsification and Quantization are the popular communication-efficient optimization approaches. Gradient sparsification in FL involves the client nodes locally selecting a subset of the values of the gradient and transmitting these with full precision. Some prominent sparsification algorithms are described below:

- **TopB Sparsification** [7], [19]: The top B gradient-values with maximum magnitude are chosen for transmission. The value of B is decided by the decision-maker using hyperparameter optimization. Sattler et al. [20] proposed

an enhanced version of TopB Sparsification with a mechanism to enable downstream compression as well as ternarization and optimal Golomb encoding of the weight updates.

- RandB Sparsification [5], [21]: In each iteration of communication, the sparsification level is chosen randomly within a range.
- Threshold-based Sparsification [8], [9]: All gradient values with a magnitude greater than a certain threshold are chosen for transmission. The value of the threshold is decided by the decision-maker using hyperparameter optimization.

Apart from gradient sparsification, gradient quantization has also been widely applied to achieve communication efficiency in FL [2], [4], [16], [17], [19], [22]–[26]. Quantization in FL is a technique in which the client nodes locally quantize the gradient to a small number of bits. In some of these proposed works, a combination of both gradient sparsification and quantization has also been proposed. In this paper, we focus on sparsification in FL. Although many sparsification methods have been applied in the FL scenario, they are based on static sparsification levels chosen before training by some hyperparameter optimization. However, there is a need for algorithms that can dynamically adapt the sparsification level for each iteration of FL based on the value of gradients and the communication costs. In this work, we propose, with theoretical guarantees, a cost-adaptive gradient sparsification algorithm with an error compensation mechanism. The proposed FL-CATE algorithm is effective for both convex and non-convex optimization.

III. PROBLEM FORMULATION

We consider Machine Learning (ML) problems involving m client nodes, each possessing local data $\mathcal{D}_i$ with $n_i = |\mathcal{D}_i|$ data points, for $i = 1, \dots, m$. Our focus centers on problems that can be formulated as follows:

$$\underset{x \in \mathbb{R}^d}{\text{minimize}} \quad f(x) := \frac{1}{n} \sum_{i=1}^m f_i(x), \quad (1)$$

where each $f_i : \mathbb{R}^d \rightarrow \mathbb{R}$ is a local loss function based on node's i data, $\mathcal{D}_i$, and $n = n_1 + \dots + n_m$. The local loss held by node i takes the form

$$f_i(x) = \frac{n_i}{n} \sum_{z \in \mathcal{D}_i} L(x; z)$$

where $L(x; z)$ is some loss function. The decision variable $x \in \mathbb{R}^d$ is the ML model that we want to learn, e.g., a deep neural network, where d is the dimension of the model. The goal is to train a global ML model $x \in \mathbb{R}^d$ with respect to the data of all nodes without the nodes needing to share their data.

We consider a FL setup in which a central server node collaborates with the client nodes to coordinate the model training. A key characteristic of FL is that no client node i needs to share its dataset $\mathcal{D}_i$ with the server node. This can be accomplished, for example, by using Federated SGD (FedSGD).

FedSGD operates by iteratively transmitting stochastic gradients from client nodes to the server, which subsequently updates a global model. More formally, the global model is initialized with some $x^0 \in \mathbb{R}^d$ by the server node. After that, at each iteration k , the server broadcasts the current global model x^k to all the clients. Each client i subsequently computes a stochastic gradient $g_i(x^k)$ based on its local data $\mathcal{D}_i$. The local stochastic gradient can be computed based on a single sample, i.e., $g_i(x^k) = \nabla L(x^k; z_i^k)$ where z_i^k is uniformly sampled data point from $\mathcal{D}_i$. Alternatively, it can be a mini-batch gradient

$$g_i(x^k) = \frac{1}{|\mathcal{D}_i^k|} \sum_{z \in \mathcal{D}_i^k} \nabla L(x^k; z),$$

where $\mathcal{D}_i^k$ is a uniformly sampled subset of $\mathcal{D}_i$. After computing the local gradient, each client node i sends its local stochastic gradient $g_i(x^k)$ to the server. The server can then perform the SGD update

$$x^{k+1} = x^k - \gamma^k \sum_{i=1}^m \frac{n_i}{n} g_i(x^k),$$

where γ^k is a learning-rate.

A drawback of FedSGD is that it involves a large amount of communications as only a single step of SGD is performed locally by clients in each iteration. In order to improve the communication efficiency, the Federated Averaging (FedAvg) algorithm has been introduced. In FedAvg, instead of one step of SGD, each client i updates a local model x_i^k by performing T local SGD steps initialized at x^k . In particular, if we let t be the local iteration index and $x_i^{k,t}$ be the local model at global iteration k and local iteration t . Then at iteration k , every client node i initializes its local iterate $x_i^{k,0} = x^k$. At each local iteration $t = 0, 1, \dots, T-1$, client node i updates its local iterate $x_i^{k,t}$:

$$x_i^{k,t+1} = x_i^{k,t} - \alpha g_i(x_i^{k,t}),$$

where $\alpha > 0$ is the local learning rate and $g_i(x_i^{k,t})$ is, again, the stochastic gradient based on random samples queried from the local dataset $\mathcal{D}_i$. Then the local model at client node i at iteration k is $x_i^k = x_i^{k,T}$. Each client node then communicates its model changes, defined as $\Delta_i^k = x_i^k - x^k$ to the server. The server then performs the Federated Averaging update

$$x^{k+1} = x^k - \gamma^k \sum_{i=1}^m \frac{n_i}{n} \Delta_i^k,$$

with learning-rate γ^t . If $T = 1$, then FedAvg is equivalent to FedSGD. But for $T > 1$ FedAvg improves the communication efficiency at the cost of more local computation.

Communication Compression. The model dimension d is often large, it is not uncommon that ML models have millions of parameters. Thus, even just transmitting one model change Δ_i^k or a gradient can be very expensive. To further improve the communication efficiency it is common to also compress the communicated information. A common compression technique is B -sparsification, which is a function $S_B : \mathbb{R}^d \rightarrow \mathbb{R}^d$ that

maps all except the B largest component of a vector $\Delta \in \mathbb{R}^d$ to zero. It can be defined component-wise as

$$[S_B(\Delta)]_j = \begin{cases} \Delta_j & \text{if } j \in I_B(\Delta) \\ 0 & \text{otherwise,} \end{cases}$$

where $I_B(\Delta)$ is the index set for the B components of Δ which are largest in magnitude. When sparsifying, a key algorithm hyperparameter that needs to be tuned is the sparsification-level B . Often many components of the communicated information Δ are redundant (small compared to the larger components), and they make lesser contribution to the model learning process. A large B value can lead to huge communication costs. On the other hand, a small B can lead to a loss of important information. Most of the existing compression techniques tune the compression level prior to running the algorithm through some hyperparameter optimization.

Error Compensation. A technique to improve the accuracy of algorithms under compression is error compensation. The idea is to correct for the error accumulation due to the compression with an error compensation step. To that end, each client node i keeps locally in memory the error accumulated until iteration k in a term e_i^k . The client nodes can then use this term to correct for the accumulation of error from compression. In particular, by performing the following modified version of FedAve

$$\begin{aligned} e_i^{k+1} &= \Delta_k^i + e_i^k - S_B(\Delta_k^i + e_i^k), \\ x^{k+1} &= x^k + \gamma^k \sum_{i=1}^n \frac{n_i}{n} S_B(\Delta_k^i + e_i^k), \end{aligned}$$

where $e_i^0 = 0$ for all client nodes i .

IV. COMMUNICATION COSTS - ENERGY, LATENCY, AND TRANSMITTED BITS

When transmitting model updates Δ , or their compressed counterparts $S_B(\Delta)$, during the execution of FL algorithms, there is always some associated communication cost. The extent of this cost naturally hinges on the dimensionality of the model $\Delta \in \mathbb{R}^d$, or d . However, if we opt for transmitting the compressed model $S_B(\Delta)$ instead, the cost becomes contingent on the compression level parameter B rather than the model's dimension d . The precise impact of this dependency is, to a large extent, contingent upon the specific communication systems we employ and the communication objectives or costs we aim to optimize.

A central idea in this paper is the adaptive adjustment of the compression level B in response to the actual data distribution and communication costs. In particular, our algorithm, presented in next section, can adapt to any communication cost. These costs can be grounded in communication standards, physical models, or even empirical measurements. In this context, we illuminate three specific communication cost considerations that frequently warrant optimization within FL systems: transmission costs, energy consumption, and latency.

A. Data Transmission Costs

In both wired and wireless communication systems, data transmission costs refer to the volume of data successfully transmitted from one node to another within a specified time frame. In the context of FL algorithms, the payload transmitted by each client node i during iteration k is the compressed model $S_B(\Delta_k^i)$. The cost of transmitting this payload is determined by the number of bits needed to encode the compressed model. Mathematically, this can be quantified as follows:

$$\text{PayLoad}(B) = B \times (\lceil \log_2(d) \rceil + \text{FPP}) \text{ bits},$$

where B is the sparsification level chosen. The $\log_2(d)$ factor is associated with indexing the B non-zero indices in the d dimensional vector $S_B(\Delta_k^i)$. FPP is the floating point precision. FPP is typically 32 for single and 64 for double precision floating-points.

In actual network communications, various types of overheads are included along with redundancies aimed at reducing the probability of errors. Additionally, there are setup costs involved in establishing and maintaining these communications [2]. Let $\text{TransmissionCost}(B)$ represent the total cost of transmitting the payload $S_B(\Delta_k^i)$ in bit, including all overheads. Thus, the transmission cost is the actual number of bits required to communicate the model when we use B -sparsification. It is often modeled using an affine model:

$$\text{TransmissionCost}(B) = C_1 \text{PayLoad}(B) + C_0,$$

where C_0 is the communication overhead while C_1 is the cost of transmitting a single unit of payload.

Often the transmitted bits depend non-linearly on B . For example, in many communication protocols, data is transmitted in the form of packets. We call this communication model as the packet model. For such communications, the transmission cost is related to the payload as follows:

$$\text{TransmissionCost}(B) = C_1 \times \left\lceil \frac{\text{PayLoad}(B)}{P_{\text{size}}} \right\rceil + C_0 \text{ bits},$$

where P_{size} is the size of a packet in bit, C_1 is the number of bits per packet including the error-correction redundancies added to the packet, C_0 is the number of overhead bits per transmission, and $\lceil \cdot \rceil$ denotes the ceiling function that rounds any real number up to the nearest integer. To find the exact values, we can use, e.g., the particular wireless protocols. For example, in the wireless standard IEEE 802.15.4, the packet model is followed, and the overhead ranges from 23-82 bytes. Since the maximum packet size is 133 bytes in this protocol, it leaves only 51-110 bytes for the payload [27].

B. Energy Costs

In systems with battery-powered nodes it is important to use the energy resources as efficiently as possible. When communicating over wireless channels, it is desirable that the compression-level is tuned so as to minimize the amount of energy consumption. The energy consumption depends on, among others, the wireless protocol and the noise in the environment. Often, an affine energy model, similar to the

one previously discussed, could provide an accurate approximation. Thus, the energy cost associated with transmitting B -sparsified vector becomes

$$\text{EnergyCost}(B) = E_1 \times \text{PayLoad}(B) + E_0 \text{ Joule.}$$

In this context, E_1 represents the energy consumption in Joules for each bit of data communicated, while E_0 denotes the baseline energy requirement for any communication, regardless of the data size. Essentially, E_1 quantifies the variable energy cost per bit, and E_0 reflects the fixed energy overhead for communication.

The values of E_1 and E_0 depend on the characteristics of the wireless channel being used. Variations in channel conditions, such as signal strength, interference, and distance, can affect how much energy is needed for transmitting data and maintaining communication. To that end, let G denote the channel power gain, ρ the transmission power of the transmitting node, and BW the allocated channel bandwidth. Then, according to the Shannon–Hartley theorem [28], the data transmission rate can be calculated as in [29]:

$$R = BW \log_2 \left(1 + \frac{G\rho}{\zeta^2} \right),$$

where ζ^2 is the additive white Gaussian noise of the transmitting device. Then $E_1 = \rho/R$ where E_0 is the communication overhead.

In many wireless protocols, the data is transmitted in the form of packets or might include redundancies to improve reliability. The energy cost of such transmissions can be modeled similarly as the packet model above, as a function of the energy cost of transmitting a single packet, the payload size, and the overhead energy-cost.

C. Latency Costs

Latency in wireless communications refers to the delay in network communication. In our FL scenario, latency shows the time taken to communicate the ML model updates between the client nodes and servers. It is often desirable to have low communication latencies in networks so as to ensure fast response times and quality of service. The cost-adaptive tuning capability of FL-CATE allows us to choose sparsification levels so as to minimize the latency in each communication round while achieving good model accuracy. The model for calculating the latency depends on the network technologies and wireless protocols. The latency can be estimated as a function of the payload size if the data is being repeatedly transmitted across the network. In this work, we quantify the latency using the ZMQ library that initiates the TCP communication once and then reuses the communication together with buffers to optimize message transmission [2].

V. FEDERATED LEARNING WITH COMMUNICATION AWARE TUNING AND ERROR COMPENSATION (FL-CATE)

We now describe FL-CATE, federated learning methods for communication aware compression-level tuning with error-correction. We also illustrate the steps of the algorithm in Fig. 1. The central idea of FL-CATE is to tune the compression

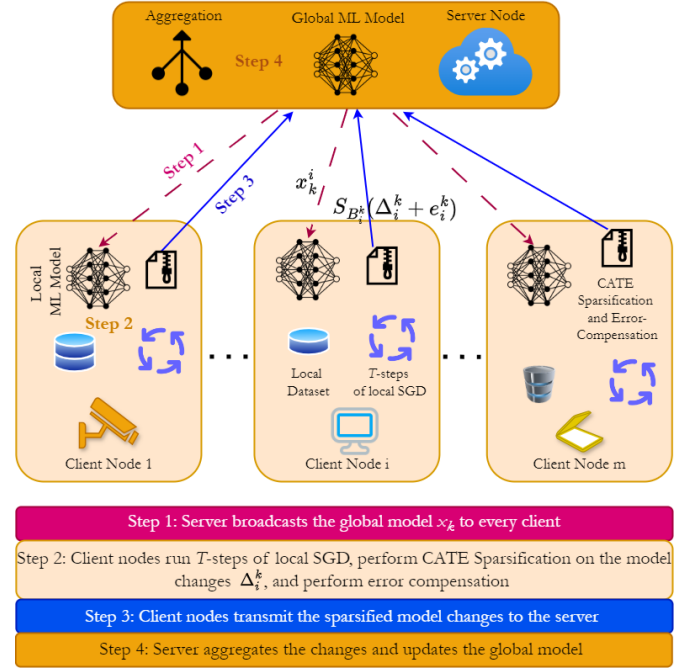


Fig. 1: Overview of the proposed FL-CATE algorithm.

level B at each client node before each transmission of model changes Δ . This requires not only adapting the compression to the transmitted information, but also to of the relevant communication costs for our system. Ideally, the appropriate compression level B is the one that optimizes the efficiency (the ratio between information and the communication cost):

$$\text{Efficiency}_B(\Delta) = \frac{\text{Information}_B(\Delta)}{\text{Cost}_B(\Delta)}. \quad (2)$$

Here $\text{Cost}_B(\Delta)$ is the communication cost used to send B model components. The communication cost depends on the cost function chosen by the decision maker and the models used by the wireless protocol (as discussed in Section IV).¹

The metric $\text{Information}_B(\Delta)$ in Eq. (2) is used to quantify the information content that the B -sparsification captures of the transmitted information Δ . As proposed in [2], the amount of information carried by the model can be quantified as:

$$\text{Information}_B(\Delta) = \frac{\|S_B(\Delta)\|^2}{\|\Delta\|^2}. \quad (3)$$

In particular, $\text{Information}_B(\Delta)$ measures the ratio between the squared norm of the output and input into the compression. Thus $\text{Information}_B(\Delta)$ measures the proportion of information captured by the compression for a given B . To illustrate this measure, consider the case where Δ is already sparse, with all except B components being zero. In this scenario, $\text{Information}_B(\Delta) = 1$, indicating that all the significant information is captured by the top B model parameters. More generally, $\text{Information}_B(\Delta)$ approaches 1 as a greater portion of Δ 's magnitude is concentrated in its

¹Though we have specified some cost functions to illustrate the ideas, in principle, any cost function can be considered. It can be based on either empirical or theoretical models.

B most significant components. On the other hand, it is easily established that [2]

$$\text{Information}_B(\Delta) \geq \frac{B}{d}$$

and $\text{Information}_B(\Delta) = \frac{B}{d}$ if and only if $\Delta = (a, a, \dots, a)$ for some $a \in \mathbb{R}$, i.e., when all the components are the same. This implies that, in the worst-case, the sparsification $S_B(\Delta)$ retains at least a fraction of the total information in Δ proportional to the ratio of the selected parameters B to the total number of parameters d . This is intuitive because $S_B(\Delta)$ consists of exactly B parts out of d of the model parameters of Δ . However, often big part of the magnitude is concentrated in the top B model parameters, meaning that $\text{Information}_B(\Delta)$ becomes close to 1.

Remark 1. Other definitions for $\text{Information}_B(\Delta)$ than the one given in Eq. (3) could also be considered, but our key idea would not change, nor will it affect the functioning of FL-CATE. This metric has been considered in this work since it is easy to implement online.

The main step in the FL-CATE algorithm involves each client node i fine-tuning their compression level B by optimizing the efficiency based on both the transmitted information Δ_i^k and with respect to the considered communication cost (Step 2 in Fig. 1). In particular, at each iteration k , each client node i adaptively tunes its compression level as follows

$$B_i^k = \underset{B \in \{1, 2, \dots, d\}}{\text{argmax}} \text{Efficiency}_B(\Delta_i^k).$$

Even though this is an integer programming problem it is a one-dimensional optimization problem, so it is solved quickly with exhaustive search. Moreover, as shown in [2], the optimization becomes submodular and can be solved quickly under linear and piece-wise linear cost functions, see Section IV.

One of the advantages of the proposed FL-CATE algorithm is that the tuning can be performed according to the user-defined cost function. Thus, the user can define the optimization goal. FL-CATE can be tuned according to the particular energy cost model used in the network, the number of bits of data communicated, the latency of communication, or any other metric. Further, this tuning is dynamic, and does not need any pre-training hyperparameter optimization, as is needed in other sparsification techniques. Moreover, the error compensation feature of FL-CATE algorithm accelerates the convergence speed.

VI. THEORETICAL ANALYSIS

In this section we provide convergence theorems for Algorithm 1. To facilitate our analysis, we impose the following standard assumptions on the objective functions.

Assumption 1. Each $f^i : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex, i.e. $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ for all $x, y \in \mathbb{R}^d$.

Assumption 2. Each $f^i : \mathbb{R}^d \rightarrow \mathbb{R}$ has L -Lipschitz continuous gradient, i.e. $\|\nabla f^i(x) - \nabla f^i(y)\| \leq L\|x - y\|$ for all $x, y \in \mathbb{R}^d$.

Algorithm 1 FL-CATE

Require: $e_i^0 = 0$ for all i , $x^0 \in \mathbb{R}^d$, the global and local step-sizes $\gamma, \alpha > 0$, the number of local steps $T > 0$, and the number of iterations $K > 0$.

```

1: for  $k = 0, 1, \dots, K - 1$  do
2:   Server broadcasts  $x_k$  to every client
3:   for Each client node  $i$  in parallel do
4:      $x_i^{k,0} = x_k$ 
5:     for  $t = 0, 1, \dots, T - 1$  do
6:        $x_i^{k,t+1} = x_i^{k,t} - \alpha g_i(x_i^{k,t})$ 
7:     end for
8:      $\Delta_i^k = x_i^k - x_i^{k,T}$ 
9:      $B_i^k = \underset{B \in [1, d]}{\text{argmax}} \text{Efficiency}_B(\Delta_i^k)$ 
10:    Transmit  $S_{B_i^k}(\Delta_i^k + e_i^k)$  to the server
11:    Update  $e_i^{k+1} = \Delta_i^k + e_i^k - S_{B_i^k}(\Delta_i^k + e_i^k)$ 
12:  end for
13:  Server performs the update:

```

$$x_{k+1} = x_k + \gamma \sum_{i=1}^n \frac{n_i}{n} S_{B_i^k}(\Delta_i^k + e_i^k)$$

```

14: end for
15: Output:  $x_K$ 

```

Assumption 3. A stochastic gradient $g^i(x)$ is unbiased and variance-bounded with respect to $\nabla f^i(x)$, i.e. $\mathbf{E}[g^i(x)] = \nabla f^i(x)$ and $\mathbf{E}\|g^i(x) - \nabla f^i(x)\|^2 \leq \sigma^2$ for all $x \in \mathbb{R}^d$.

Assumptions 1-3 are standard for establishing bounds on the convergence rate of stochastic optimization algorithms. We establish convergence both for convex and non-convex problems. Assumption 1 ensures convexity, which allows us to obtain stronger convergence rates. Examples of convex models include linear regression, logistic regression, and support vector machines. However, our analysis extends its reach beyond convexity to embrace non-convex problems, as encountered in deep learning scenarios. Meanwhile, Assumption 3 assures that the gradients remain unbiased and possess finite variance throughout the optimization process. This assumption is valid in the context of Algorithm 1, as it relies on batched stochastic gradients obtained through uniform sampling from a finite dataset.

The first result is the convergence for this algorithm on convex optimization.

Theorem 1 (Convex optimization). *Consider the iterates $\{x_k\}_{k \in \mathbb{N}}$ generated by Algorithm 1 for Problem (1) where Assumptions 1, 2, and 3 hold. Let the step-size $\gamma = \bar{\gamma}/T$, where*

$$0 < \bar{\gamma} \leq \frac{1}{L} \cdot \min \left(\frac{1}{10}, \frac{1}{8[\Gamma + 1]} \right).$$

Then,

$$\mathbf{E} \left[f \left(\frac{1}{K} \sum_{k=0}^{K-1} x_k \right) - f(x^*) \right] \leq \frac{2\|x_0 - x^*\|^2}{\bar{\gamma}K} + 2\bar{\gamma}e,$$

where $e = \beta_2 \sigma^2 + (2\beta_1/n) \sum_{i=1}^n \|\nabla f^i(x^*)\|^2$, $\beta_1 = \Gamma + 1$, $\beta_2 = \Gamma + 4$, and $\Gamma = (12/\alpha)(1 + 2/\alpha)(1 - \alpha)$, where $\alpha = \min_{i,k} B_i^k$.

Proof. See Appendix B. $\square$

Theorem 1 establishes the convergence of Algorithm 1 for convex optimization. The step-size and convergence bound of Algorithm 1 depend on the number of local steps T , the Lipschitz constant L , and α . Note that we always have the following lower bound $\alpha \geq 1/d$ on α , since $B_i^k \gg 1$ for all i and k . Furthermore, from Theorem 1, Algorithm 1 enjoys the $\mathcal{O}(1/\sqrt{K})$ convergence towards the exact optimal solution when we let $\bar{\gamma} = \mathcal{O}(1/\sqrt{K})$. By letting $\bar{\gamma} = A/\sqrt{K}$ where $A \leq \frac{1}{L} \cdot \min\left(\frac{1}{10}, \frac{1}{8[\Gamma+1]}\right)$ into Theorem 1, Algorithm 1 satisfies:

$$\mathbb{E} \left[f \left(\frac{1}{K} \sum_{k=0}^{K-1} x_k \right) - f(x^*) \right] \leq \frac{2\|x_0 - x^*\|^2/A + 2Ae}{\sqrt{K}}.$$

We next establish the convergence for Algorithm 1 for non-convex optimization. To that end, we need to make an additional assumption on data-Heterogeneity.

Assumption 4. Each $f^i : \mathbb{R}^d \rightarrow \mathbb{R}$ is ξ -data heterogeneous with respect to f , i.e. $\|\nabla f^i(x) - \nabla f(x)\|^2 \leq \xi^2$ for all $x \in \mathbb{R}^d$.

This assumption is standard in FL, and limits how different the data distributions are between client nodes. Under this assumption we get the following results.

Theorem 2 (Non-convex optimization). *Consider the iterates $\{x_k\}_{k \in \mathbb{N}}$ generated by Algorithm 1 for Problem (1) where Assumptions 1-4 hold. Let $f_{\inf} = \inf_{x \in \mathbb{R}^d} f(x)$, let the step-size $\gamma = \bar{\gamma}/T$, where*

$$0 < \bar{\gamma} \leq \frac{1}{L} \cdot \min \left(\frac{1}{12}, \frac{1}{8\sqrt{2}} \sqrt{\frac{\alpha}{\Gamma}} \right),$$

and let $\hat{x}_K$ be selected uniformly at random from $\{x_0, x_1, \dots, x_{K-1}\}$. Then,

$$\mathbb{E} \|\nabla f(\hat{x}_K)\|^2 \leq \frac{64[f(x_0) - f_{\inf}]}{\bar{\gamma}K} + 64\bar{\gamma}\bar{e},$$

where $\bar{e} = (33L/16 + L\Gamma/[12\alpha])\sigma^2 + (9/8 + L\Gamma/[6\alpha])\xi^2$, and $\Gamma = 6(1 + 2/\alpha)(1 - \alpha)$ where $\alpha = \min_{i,k} B_i^k$.

Proof. See Appendix C. $\square$

Analogously to Theorem 1, Theorem 2 states the sub-linear convergence for Algorithm 1 towards the solution with the residual error for non-convex optimization. Unlike Theorem 1, Theorem 2 requires the data heterogeneity condition (Assumption 4) to prove the result. Furthermore, from this theorem, Algorithm 1 enjoys the $\mathcal{O}(1/\sqrt{K})$ rate when $\gamma = \mathcal{O}(1/\sqrt{K})$, which is a similar rate obtained from [17] without requiring the unbiased property of compressors. We can show this by setting $\bar{\gamma} = B/\sqrt{K}$ where $B \leq \frac{1}{L} \cdot \min\left(\frac{1}{12}, \frac{1}{8\sqrt{2}} \sqrt{\frac{\alpha}{\Gamma}}\right)$ into Theorem 2:

$$\mathbb{E} \|\nabla f(\hat{x}_K)\|^2 \leq \frac{64[f(x_0) - f^*]/B + 64B\bar{e}}{\sqrt{K}}.$$

The parameter ξ , as outlined in Assumption 1, serves to bound the data heterogeneity, influencing the convergence properties of the algorithm. However, it is important to note that ξ does not need to be explicitly known to run the algorithm, such as when tuning the step-size. The heterogeneity-parameter ξ does, however, influence the convergence, specifically, as ξ grows larger the convergence bound becomes less tight. Therefore, the parameter ξ well captures the intuition that as the level of heterogeneity becomes larger, the training process becomes more challenging.

VII. NUMERICAL RESULTS

We now evaluate and compare the efficiency of the proposed FL-CATE algorithm with three baseline schemes: 1) without Sparsification (FL-WS), 2) with Threshold-Based Sparsification and Error compensation (FL-TBSE), 3) with Constant- T Sparsification and Error compensation (FL-CTSE). It should be noted that there is absence of a good scheme for tuning the threshold in FL-TBSE and the sparsification level (B) in FL-CTSE. In all our experiments, a grid search was performed for finding the optimal B for FL-CTSE and the optimal threshold for FL-TBSE so as to achieve the desired accuracy.² We have only illustrated the results for the optimal tuning parameters for FL-CTSE and FL-TBSE. It is worth highlighting that the proposed FL-CATE algorithm automatically performs the optimal tuning without the need of such hyperparameter optimization. This is a significant advantage over other methods.

A. The Simulation Setup

We conduct nine experiments under diverse network conditions as shown in the table I. All the experiments are performed on the Windows operating system using the 11th Gen Intel(R) Core(TM) i7-1185G7 CPU @ 3.00GHz with 32 GB RAM. We consider a network of five client nodes that train a common ML model in collaboration with one cloud server. Experiments 1-6 use the MNIST Handwritten Digit Classification Dataset [30]. In experiments 7-9, a CNN model is trained using the Fashion-MNIST Dataset [31]. Both these datasets have 60,000 training images and 10,000 testing images.

Each experiment was run five times, and results were averaged. In each run-time, the dataset was randomly distributed to the nodes. The testing dataset was the same for all simulations and never used for training. In the MLP model, we have two hidden layers of size 700 and 500. The learning rates for the LR and MLP models are 0.1 and 0.001, respectively. For CNN model, the architecture is same as in [32] with over 400,000 weights. For the packet model, we consider wireless standard IEEE 802.15.4, in which the packet size is around 133 Bytes and the communication overhead per packet is around 30 Bytes [27]. Other simulation parameters are summarized in table II. To verify the robustness of the proposed approach, we have added a noise in cost calculations with SNR (Signal to Noise Ratio) varying in the range of 30 – 50.

²For FL-TBSE, the grid search was performed for the values 0.1, 0.01, 0.001, 0.001, and for FL-CTSE, the grid search was performed for the values 0.001, 0.01, 0.02, 0.03, 0.04, 0.05, 0.1. The rationale used for choosing these values was to choose values of different mathematical orders, and then to choose different values of the best-performing mathematical order.

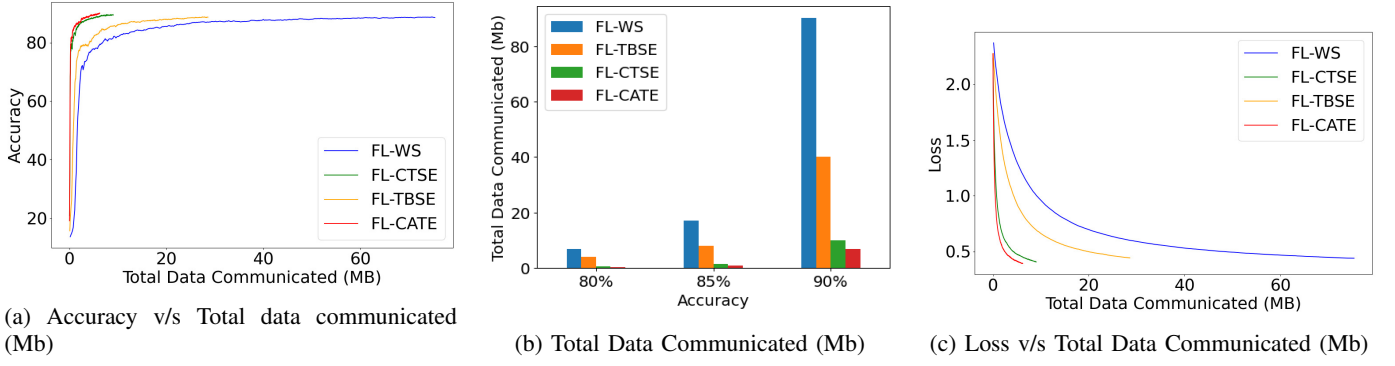


Fig. 2: These are the results of **Experiment 1** of performing Federated Learning (with error compensation) on the MNIST dataset to train a **logistic regression model**. The cost to be minimized is the number of **bits** communicated.

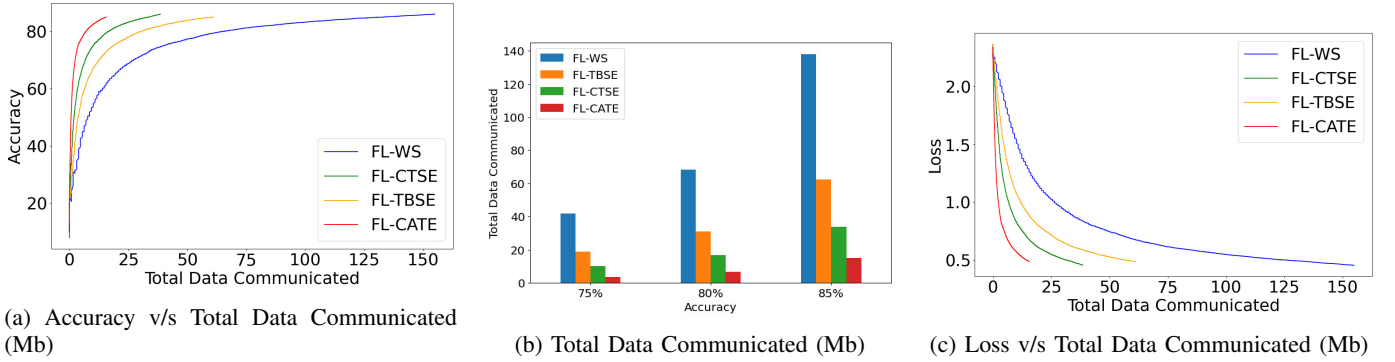


Fig. 3: These are the results of **Experiment 2** of performing Federated Learning (with error compensation) on the MNIST dataset to train a **Multilayer Perceptron model**. The cost to be minimized is the number of **bits** communicated.

Exp.	Cost	Model	Loss fun. Type	Comm. Model
1	Bits	LR	Convex	Affine Model
2	Bits	MLP	Non-Convex	Affine Model
3	Energy	LR	Convex	Packet Model
4	Energy	MLP	Non-Convex	Packet Model
5	Time	LR	Convex	Affine Model
6	Time	MLP	Non-Convex	Affine Model
7	Bits	CNN	Non-Convex	Packet Model
8	Energy	CNN	Non-Convex	Packet Model
9	Time	CNN	Non-Convex	Packet Model

TABLE I: Experiments with their specifications. The first column is the experiment number. The second column specifies the communication cost considered to adaptively tune the sparsification level. The third one is the ML model that was trained in the experiment. LR stands for Linear Regression, and MLP stands for Multilayer Perceptron. The fourth column specifies the type of loss function associated with training the ML model. The fifth column in the table specifies the kind of communication model considered during the simulations.

Simulation Parameters			
Parameter	Value	Parameter	Value
T	5 steps	e_1	$1\mu J/bit$
l_1	3.1 ns/bit	R	$\times 300$
BW^d	20MHz	m	5
BW^u	20MHz	f	0.5 GHz

TABLE II: The different simulation parameters chosen for the experiments.

B. Cost Functions 1: Communicated Bits

Experiments 1 and 2 were conducted, focusing on minimizing the number of bits communicated while achieving good accuracy for the model. The results of experiments 1 and 2 are plotted in Fig. 2 and Fig. 3, respectively. Experiment 1 comprised training an LR model (convex optimization), while Experiment 2 comprised training an MLP model (non-convex optimization). Fig. 2b and Fig. 3b show the bar plots to compare the total data communicated during the complete FL process. Fig. 2b shows the data communicated to achieve an accuracy of 80%, 85%, and 90%. Fig. 3b shows the data communicated to achieve an accuracy of 75%, 80%, and 85%. We can see from Fig. 2b that on average, for convex optimization, the amount of data communicated in FL-CATE is lesser than FL-WS, FL-TBSE and FL-CTSE by 92.7%, 83.93.2%, and 31.5% respectively. Similarly, it is evident from Fig. 3b that for non-convex optimization, on average, the amount of data communicated in FL-CATE is lesser than FL-WS, FL-TBSE and FL-CTSE by 87.5%, 72% and 54.6% respectively. Noticeably, FL-CATE communicates much less data while achieving the same accuracy as other baselines.

FL-CATE outperforms the other baselines during the whole training process - for both convex and non-convex optimization. This is evident from Fig. 2a and Fig. 3a. The shapes of curves of all baselines are similar, the only difference being the magnitude of data communicated. FL-CTSE is the baseline that performs the closest, although lesser than

FL-CATE. However, the results of FL-CTSE plotted herein were the best results obtained after intensive hyperparameter optimization, whereas FL-CATE achieves these without any hyper-parameter optimization. The contrast between the results of convex and non-convex optimization is that convex optimization involved lesser communication of data than non-convex because of the simplicity and smaller size of the ML model. Also, for the non-convex model (MLP), we terminate the experiment when an accuracy of 85% is reached, as opposed to 90% in the convex model (logistic regression). Again, this is because of the high memory requirements and slower convergence in the more complex model.

Fig. 2c and Fig. 3c depict the loss v/s bits for convex and non-convex optimization, respectively. In Fig. 2c, FL-CATE performs best by achieving the minimum loss of 0.22, while none of the other baselines achieve this level of loss. When we compare the data communicated by the algorithms to achieve a common loss of 0.5, it is clear that FL-CATE achieves convergence by communicating just 2.5 Mb of data. However, to achieve the same loss, FL-CTSE, FL-TBSE, and FL-WS had to communicate 4 Mb, 22 Mb, and 55 Mb, respectively. Also, FL-CATE adaptively tunes the compression level, while other algorithms must be tuned beforehand to achieve good performance. The graphs plotted in this work for other baselines are plotted for the best performance they could achieve after extensive tuning experiments. Similar convergence trends were seen for non-convex optimization (Fig. 3c).

C. Cost Function 2: Transmitted Energy

Experiments 3 and 4 were conducted, focusing on minimizing the amount of energy consumed while achieving good accuracy for the model. The results of experiments 3 and 4 are plotted in Fig. 4 and Fig. 5, respectively. Experiment 3 corresponds to training an LR model, while Experiment 4 to training an MLP model. Fig. 4b and Fig. 5b show the bar plots to compare the total energy consumed during the complete FL process. Fig. 4b shows the energy consumed to achieve an accuracy of 80%, 85%, and 90%. Fig. 5b shows the data communicated to achieve an accuracy of 75%, 80%, and 85%. We can see from Fig. 4b that on average, for convex optimization, the amount of energy consumed in FL-CATE is lesser than FL-WS, FL-TBSE and FL-CTSE by 91.3%, 82.7%, and 55% respectively. Similarly, it is evident from Fig. 5b that for non-convex optimization, on average, the amount of energy consumed in FL-CATE is lesser than FL-WS, FL-TBSE and FL-CTSE by 89%, 73% and 30% respectively. Noticeably, FL-CATE leads to substantially less amount of energy consumption while achieving the same accuracy as other baselines. This makes FL-CATE especially suited for FL in IoT networks, which are characterized by IoT devices with limited energy resources.

FL-CATE outperforms the other baselines during the whole training process. This is clear from Fig. 4a and Fig. 5a. The shapes of the curves of all baselines are similar, the only difference being the amount of energy consumed. The contrast between the results of convex and non-convex optimization is that convex optimization involved lesser energy consumption than non-convex because of the smaller size of the ML model.

Fig. 4c and Fig. 5c depict the loss v/s energy plots for convex and non-convex optimization, respectively. In Fig. 4c, FL-CATE performs best by achieving the minimum loss of 0.23, while none of the other baselines achieve this loss level. When we compare the amount of energy consumed by the algorithms to achieve a common loss of 0.5, it is clear that FL-CATE converges with an energy cost of around 0.15 MJ. However, to achieve the same loss, FL-CTSE, FL-TBSE, and FL-WS resulted in an energy consumption of 0.45 MJ, 1 MJ, and 3 MJ, respectively. Also, FL-CATE adaptively tunes the compression level, while other algorithms must be tuned beforehand to achieve good performance. Similar convergence trends were seen for non-convex optimization (Fig. 5c).

D. Cost Function 3: Latency

Experiments 5 and 6, shown in Fig. 6 and Fig. 7 respectively, focus on minimizing the communication time while achieving good accuracy for the model. Fig. 6b and Fig. 7b show the bar plots to compare the total latency associated with the complete FL process. Fig. 6b shows the communication time needed to achieve an accuracy of 80%, 85%, and 90%. Fig. 7b shows the latency involved to achieve an accuracy of 75%, 80%, and 85%. We can see from Fig. 6b that on average, for convex optimization, the communication time in FL-CATE is lesser than FL-WS, FL-TBSE and FL-CTSE by 94.2%, 85.85%, and 45.5% respectively. Similarly, it is evident from Fig. 7b that for non-convex optimization, on average, the communication time in FL-CATE is lesser than FL-WS, FL-TBSE and FL-CTSE by 89.5%, 77.7% and 58.1% respectively. Similar to the other two costs considered earlier, FL-CATE involves substantially less amount of communication time while achieving the same accuracy as other baselines.

FL-CATE outperforms the other baselines during the whole training process. This is evident from Fig. 6a and Fig. 7a. The shapes of the curves of all baselines are similar, the only difference being the amount communication time. For the non-convex model (MLP), the communication time seems to be lesser than for convex model, although the amount of data communicated is much larger. This is because we terminate the latter experiment when an accuracy of 85% is reached, as opposed to 90% in the convex model (logistic regression).

Fig. 6c and Fig. 7c depict the loss v/s time plots for convex and non-convex optimization, respectively. In Fig. 4c, FL-CATE performs best by achieving the minimum loss of 0.19, while none of the other baselines achieve this loss level. When we compare the amount of communication time before achieving a common loss of 0.5, it is clear that FL-CATE achieves convergence with a communication time of 80 ms. However, to achieve the same loss, FL-CTSE, FL-TBSE, and FL-WS involved a communication time of 160 ms, 500 ms, and 1050 ms, respectively. Also, FL-CATE adaptively tunes the compression level, while other algorithms must be tuned beforehand to achieve good performance. Similar convergence trends were seen for non-convex optimization (Fig. 7c). The future IoT networks will be characterized by highly delay-sensitive applications. The speed-up achieved by FL-CATE by reducing the communication time makes it especially suitable for several delay-sensitive applications.

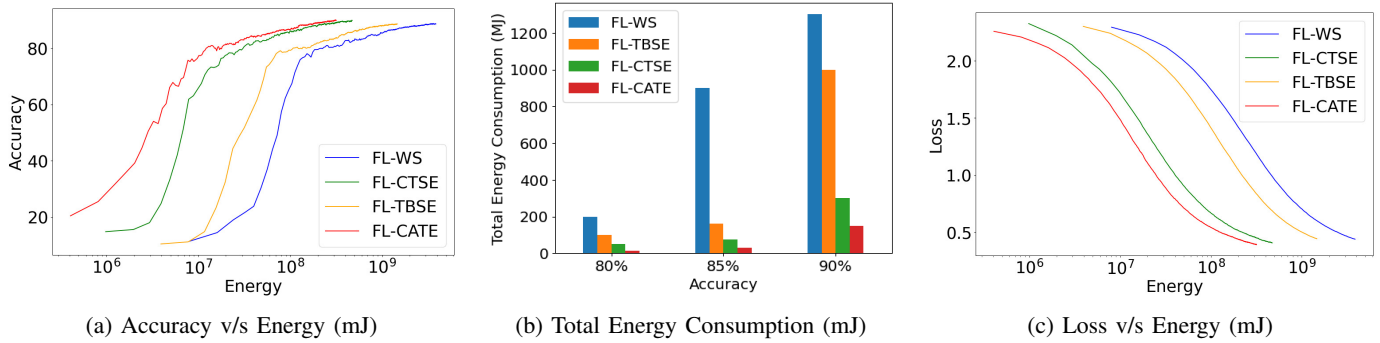


Fig. 4: These are the results of **Experiment 3** of performing Federated Learning (with error compensation) on the MNIST dataset to train a **logistic regression model**. The cost to be minimized is the amount of **energy** consumed.

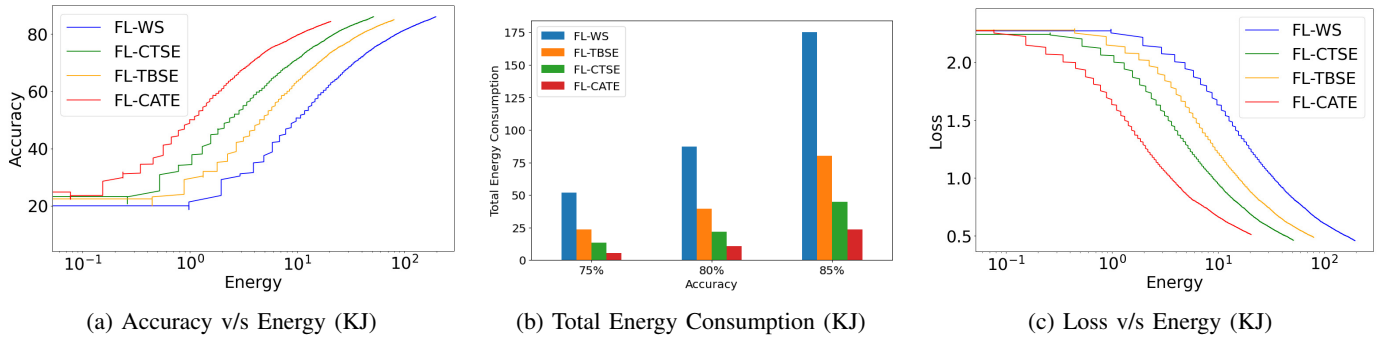


Fig. 5: These are the results of **Experiment 4** of performing Federated Learning (with error compensation) on the MNIST dataset to train a **Multilayer Perceptron model**. The cost to be minimized is the amount of **energy** consumed.

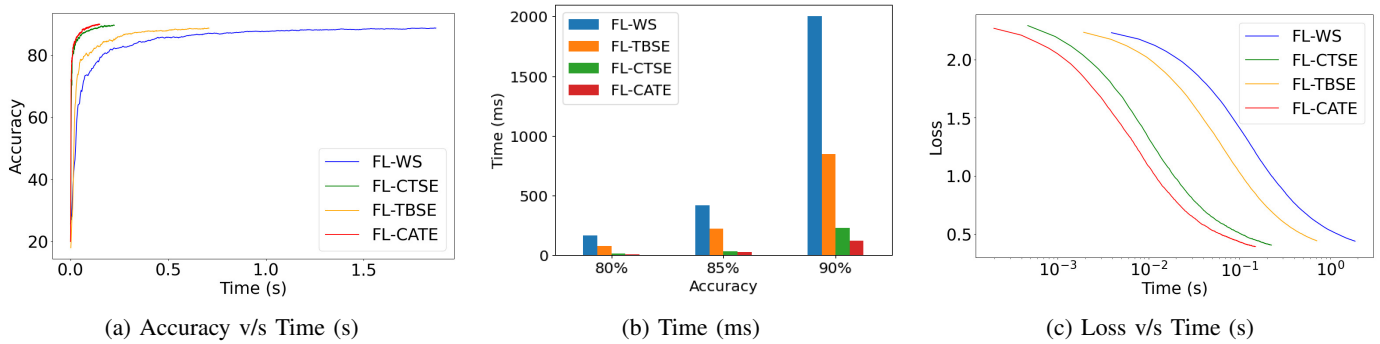


Fig. 6: These are the results of **Experiment 5** of performing Federated Learning (with error compensation) on the MNIST dataset to train a **logistic regression model**. The cost to be minimized is the total **communication time**.

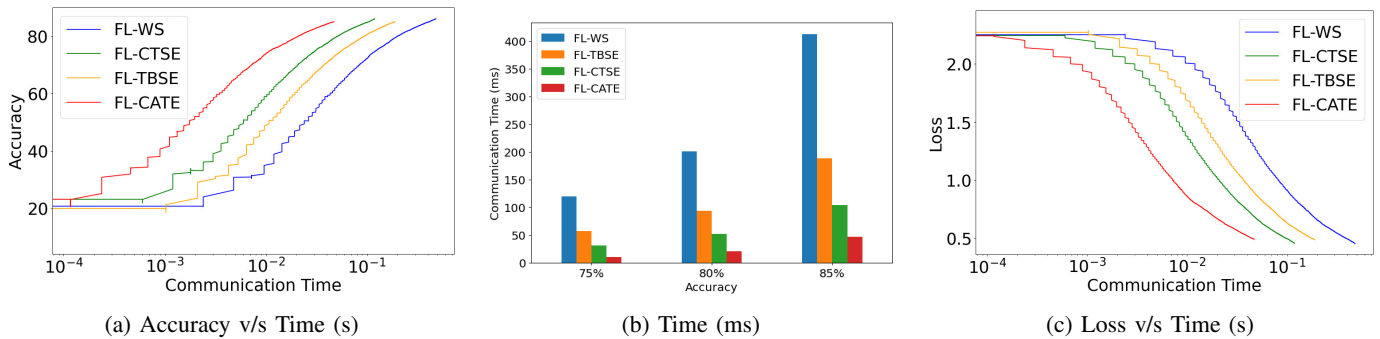


Fig. 7: These are the results of **Experiment 6** of performing Federated Learning (with error compensation) on the MNIST dataset to train an **MLP model**. The cost to be minimized is the total **communication time**.

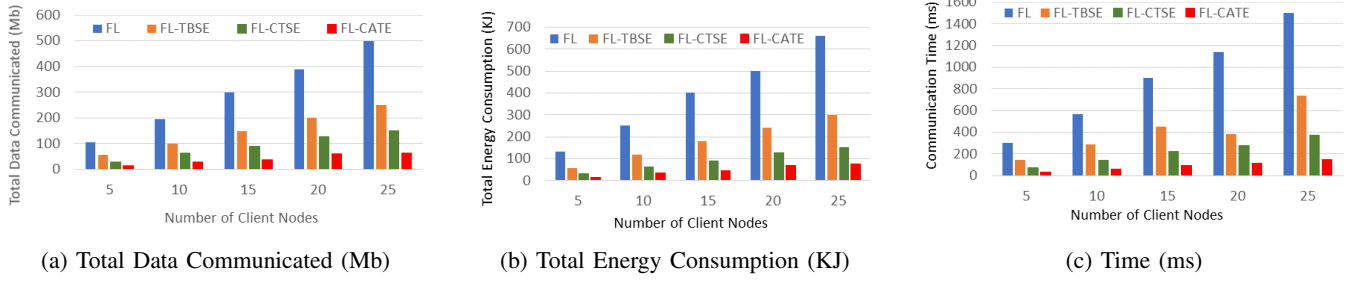


Fig. 8: These are the results of **Experiment 7, 8, 9** of performing Federated Learning (with error compensation) on the Fashion-MNIST dataset to train a **CNN model**.

E. Scalability and Generalizability

Experiments 7, 8, and 9 have been conducted to verify the scalability and generalizability of the proposed approach. The scalability is verified by varying the number of client nodes in the network from 5 to 25. Generalizability is verified by training another non-convex ML model (CNN) on a different dataset (Fashion-MNIST dataset). Fig. 8 shows the results of these experiments. Fig. 8a shows the results when the focus is on minimizing the number of bits communicated, whereas Fig. 8b and Fig. 8c correspond to the costs of energy consumption and latency respectively. It is clearly evident that despite varying numbers of client nodes, different ML modes, and noise in communication, the proposed FL-CATE algorithm outperforms the other baselines by significant proportions.

VIII. CONCLUSION

We propose the use of CAT sparsification framework with error compensation to adapt the compression level based on the communication cost in each iteration of FL. We provide theoretical convergence guarantees of the proposed FL-CATE algorithm for both convex and non-convex optimization. Six extensive experiments were performed to verify the results on two ML models (convex and non-convex) and three cost functions (bits transmitted, latency, energy). Simulation results show that FL-CATE achieves the fastest convergence and best accuracy, as compared to the other three baseline algorithms FL-WS, FL-TBSE, and FL-CTSE. Also, FL-CATE algorithm achieves these results while involving substantially less amount of communication cost. Moreover, FL-CATE algorithm does not need to be tuned like other algorithms for different communication costs and wireless protocols. These results are obtained because of the intelligent cost-dependent adaptation of sparsification level in each iteration of FL, while simultaneously compensating for the errors iteratively.

APPENDIX A PRELIMINARIES FOR PROOFS

The update for Algorithm 1 with $\gamma = \bar{\gamma}/T$ can be rewritten into:

$$z_{k+1} = z_k - \bar{\gamma}v_k, \quad \text{where } v_k = \frac{1}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} g^i(x_{k,l}^i), \quad (4)$$

and $z_k = x_k + (1/n) \sum_{i=1}^n e_k^i$. From Algorithm 1 with $\gamma = \bar{\gamma}/T$, we have

$$x_{k,l}^i = \begin{cases} x_{k,0}^i - \frac{\bar{\gamma}}{T} \sum_{j=0}^{l-1} g^j(x_{k,j}^i) & \text{for } l = 1, 2, \dots, T \\ x_k & \text{for } l = 0 \end{cases} \quad (5)$$

Next we introduce useful inequalities for analyzing Algorithm 1. For $\theta > 0$ and $x, y, x_1, x_2, \dots, x_N \in \mathbb{R}^d$,

$$\left\| \sum_{i=1}^N x_i \right\|^2 \leq N \sum_{i=1}^N \|x_i\|^2, \quad (6)$$

$$\langle x, y \rangle \leq (\theta/2)\|x\|^2 + \|y\|^2/(2\theta), \quad (7)$$

$$2\langle x, y \rangle = \|x\|^2 + \|y\|^2 - \|x - y\|^2, \quad \text{and} \quad (8)$$

$$\|x + y\|^2 \leq (1 + \theta)\|x\|^2 + (1 + 1/\theta)\|y\|^2. \quad (9)$$

We will also use the next lemma, which follows directly from Lemma 1 in [33].

Lemma 1. *The compression S_{B^k} in Algorithm 1 is α -contractive for all i and k , i.e. $\|\hat{S}_{B(i,k)}(x) - x\|^2 \leq (1 - \alpha)\|x\|^2$ for all $x \in \mathbb{R}^d$.*

Next, we present two lemmas which are instrumental to the analysis. The first lemma bounds the Euclidean distance between the local iterate $x_{k,l}^i$ and the global iterate x_k , while the second lemma bounds the Euclidean norm of the compression error e_k^i .

Lemma 2. *Consider the iterates $\{x_k\}_{k \in \mathbb{N}}$ generated by Algorithm 1 for Problem (1) where Assumptions 2 and 3 hold. Let $\bar{\gamma} \leq 1/(\sqrt{6}L)$. Then,*

$$\frac{1}{T} \sum_{j=0}^{T-1} \mathbf{E} \|x_{k,j}^i - x_k\|^2 \leq 6\bar{\gamma}^2 \mathbf{E} \|\nabla f^i(x_k)\|^2 + 6\bar{\gamma}^2 \sigma^2. \quad (10)$$

Proof. From the definition of the Euclidean norm, by (5), and by (6)

$$\|x_{k,l}^i - x_k\|^2 \leq \frac{\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|g^j(x_{k,j}^i)\|^2.$$

Next, by the fact that $\|x + y + z\|^2 \leq 3\|x\|^2 + 3\|y\|^2 + 3\|z\|^2$ with $x = g^i(x_{k,j}^i) - \nabla f^i(x_{k,j}^i)$, $y = \nabla f^i(x_{k,j}^i) - \nabla f^i(x_k)$,

and $z = \nabla f^i(x_k)$,

$$\begin{aligned} \|x_{k,l}^i - x_k\|^2 &\leq \frac{3\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|g^i(x_{k,j}^i) - \nabla f^i(x_{k,j}^i)\|^2 \\ &\quad + \frac{3\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|\nabla f^i(x_{k,j}^i) - \nabla f^i(x_k)\|^2 \\ &\quad + \frac{3\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|\nabla f^i(x_k)\|^2. \end{aligned}$$

By Assumption 2 (Lipschitz continuity of $\nabla f_i(x)$),

$$\begin{aligned} \|x_{k,l}^i - x_k\|^2 &\leq \frac{3\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|g^i(x_{k,j}^i) - \nabla f^i(x_{k,j}^i)\|^2 \\ &\quad + \frac{3L^2\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|x_{k,j}^i - x_k\|^2 \\ &\quad + \frac{3\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|\nabla f^i(x_k)\|^2. \end{aligned}$$

By taking the expectation, and by Assumption 3 (bounded variance),

$$\begin{aligned} \mathbf{E}\|x_{k,l}^i - x_k\|^2 &\leq \frac{3L^2\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \mathbf{E}\|x_{k,j}^i - x_k\|^2 \\ &\quad + 3\bar{\gamma}^2 \mathbf{E}\|\nabla f^i(x_k)\|^2 + 3\bar{\gamma}^2 \sigma^2. \end{aligned}$$

If $\bar{\gamma} \leq 1/(\sqrt{6}L)$, then

$$\frac{1}{T} \sum_{j=0}^{T-1} \mathbf{E}\|x_{k,j}^i - x_k\|^2 \leq 6\bar{\gamma}^2 \mathbf{E}\|\nabla f^i(x_k)\|^2 + 6\bar{\gamma}^2 \sigma^2.$$

□

Lemma 3. Consider the iterates $\{x_k\}_{k \in \mathbb{N}}$ generated by Algorithm 1 for Problem (1) where Assumptions 2, 3 hold. Let $\bar{\gamma} \leq 1/(\sqrt{6}L)$. Then,

$$\begin{aligned} \mathbf{E}\|e_{k+1}^i\|^2 &\leq (1 - \alpha/2) \mathbf{E}\|e_k^i\|^2 \\ &\quad + 6(1 + 2/\alpha)(1 - \alpha)\bar{\gamma}^2 [\mathbf{E}\|\nabla f^i(x_k)\|^2 + \sigma^2]. \end{aligned} \quad (11)$$

Proof. From the definition of the Euclidean norm and e_{k+1}^i , and by Lemma 1, we have for $\theta > 0$

$$\begin{aligned} \|e_{k+1}^i\|^2 &\leq (1 - \alpha) \|x_{k,T}^i - x_k + e_k^i\|^2 \\ &\stackrel{(9)}{\leq} (1 + \theta)(1 - \alpha) \|e_k^i\|^2 + (1 + 1/\theta)(1 - \alpha) \|x_{k,T}^i - x_k\|^2. \end{aligned}$$

If $\theta = \alpha/2$, then $(1 + \alpha/2)(1 - \alpha) \leq 1 - \alpha/2$ and by taking the expectation

$$\mathbf{E}\|e_{k+1}^i\|^2 \leq (1 - \alpha/2) \mathbf{E}\|e_k^i\|^2 + (1 + 2/\alpha)(1 - \alpha) \mathbf{E}\|x_{k,T}^i - x_k\|^2.$$

To bound $\|e_{k+1}^i\|^2$, we bound $\|x_{k,T}^i - x_k\|^2$. By (5), and (6),

$$\|x_{k,T}^i - x_k\|^2 \leq \frac{\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|g^i(x_{k,j}^i)\|^2.$$

By the fact that $\|x + y + z\|^2 \leq 3\|x\|^2 + 3\|y\|^2 + 3\|z\|^2$ with $x = g^i(x_{k,j}^i) - \nabla f^i(x_{k,j}^i)$, $y = \nabla f^i(x_{k,j}^i) - \nabla f^i(x_k)$, and $z = \nabla f^i(x_k)$,

$$\begin{aligned} \|x_{k,T}^i - x_k\|^2 &\leq \frac{3\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|g^i(x_{k,j}^i) - \nabla f^i(x_{k,j}^i)\|^2 \\ &\quad + \frac{3\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|\nabla f^i(x_{k,j}^i) - \nabla f^i(x_k)\|^2 \\ &\quad + \frac{3\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|\nabla f^i(x_k)\|^2. \end{aligned}$$

By Assumption 2 (Lipschitz continuity of $\nabla f_i(x)$),

$$\begin{aligned} \|x_{k,T}^i - x_k\|^2 &\leq \frac{3\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|g^i(x_{k,j}^i) - \nabla f^i(x_{k,j}^i)\|^2 \\ &\quad + \frac{3L^2\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|x_{k,j}^i - x_k\|^2 \\ &\quad + \frac{3\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \|\nabla f^i(x_k)\|^2. \end{aligned}$$

Next, by taking the expectation, and by Assumption 3,

$$\begin{aligned} \mathbf{E}\|x_{k,l}^i - x_k\|^2 &\leq \frac{3L^2\bar{\gamma}^2}{T} \sum_{j=0}^{T-1} \mathbf{E}\|x_{k,j}^i - x_k\|^2 \\ &\quad + 3\bar{\gamma}^2 \mathbf{E}\|\nabla f^i(x_k)\|^2 + 3\bar{\gamma}^2 \sigma^2. \end{aligned}$$

Next, by (10) from Lemma 2

$$\mathbf{E}\|x_{k,l}^i - x_k\|^2 \leq (18L^2\bar{\gamma}^4 + 3\bar{\gamma}^2) [\mathbf{E}\|\nabla f^i(x_k)\|^2 + \sigma^2].$$

Since $\bar{\gamma} \leq 1/(\sqrt{6}L)$, we get

$$\mathbf{E}\|x_{k,l}^i - x_k\|^2 \leq 6\bar{\gamma}^2 [\mathbf{E}\|\nabla f^i(x_k)\|^2 + \sigma^2].$$

Finally, plugging the upper-bound of $\mathbf{E}\|x_{k,l}^i - x_k\|^2$ into the upper-bound of $\mathbf{E}\|e_{k+1}^i\|^2$ yields

$$\begin{aligned} \mathbf{E}\|e_{k+1}^i\|^2 &\leq (1 - \alpha/2) \mathbf{E}\|e_k^i\|^2 \\ &\quad + 6(1 + 2/\alpha)(1 - \alpha)\bar{\gamma}^2 [\mathbf{E}\|\nabla f^i(x_k)\|^2 + \sigma^2]. \end{aligned}$$

□

APPENDIX B PROOF OF THEOREM 1

We prove the main result in five steps.

Step 1) Bound $\mathbf{E}\|z_{k+1} - x^\|^2$:* From the definition of the Euclidean norm and by (4),

$$\|z_{k+1} - x^*\|^2 = \|z_k - x^*\|^2 - 2\bar{\gamma} \langle v_k, z_k - x^* \rangle + \bar{\gamma}^2 \|v_k\|^2.$$

Next, by taking the expectation and by Assumption 3 (unbiased property of stochastic gradients), i.e. $\mathbf{E}[v_k] = \bar{v}_k := (1/[nT]) \sum_{i=1}^n \sum_{l=0}^{T-1} \nabla f^i(x_{k,l}^i)$,

$$\begin{aligned} \mathbf{E}\|z_{k+1} - x^*\|^2 &= \mathbf{E}\|z_k - x^*\|^2 - 2\bar{\gamma} \mathbf{E} \langle \bar{v}_k, z_k - x^* \rangle \\ &\quad + \bar{\gamma}^2 \mathbf{E}\|v_k\|^2. \end{aligned}$$

By the fact that $\|x + y + z\|^2 \leq 3\|x\|^2 + 3\|y\|^2 + 3\|z\|^2$ with $x = \nabla f(x_k)$, $y = \bar{v}_k - \nabla f(x_k)$, and $z = v_k - \bar{v}_k$,

$$\mathbf{E}\|z_{k+1} - x^*\|^2 \leq \mathbf{E}\|z_k - x^*\|^2 - 2\bar{\gamma}\mathbf{E}\langle \bar{v}_k, z_k - x^* \rangle + 3\bar{\gamma}^2\mathbf{E}\|\nabla f(x_k)\|^2 + 3\bar{\gamma}^2(T_1 + T_2),$$

where $T_1 = \mathbf{E}\|\bar{v}_k - \nabla f(x_k)\|^2$, and $T_2 = \mathbf{E}\|v_k - \bar{v}_k\|^2$. Since

$$\begin{aligned} 2\bar{\gamma}\langle \bar{v}_k, z_k - x^* \rangle &= 2\bar{\gamma}\langle \bar{v}_k, x_k - x^* \rangle + 2\bar{\gamma}\langle \bar{v}_k, z_k - x_k \rangle \\ &\stackrel{(7)}{\leq} 2\bar{\gamma}\langle \bar{v}_k, x_k - x^* \rangle + \gamma^2\|\bar{v}_k\|^2 + \|x_k - z_k\|^2 \\ &\stackrel{(6)}{\leq} 2\bar{\gamma}\langle \bar{v}_k, x_k - x^* \rangle + 2\gamma^2\|\nabla f(x_k)\|^2 \\ &\quad + 2\gamma^2T_1 + \|x_k - z_k\|^2, \end{aligned}$$

we get

$$\mathbf{E}\|z_{k+1} - x^*\|^2 \leq \mathbf{E}\|z_k - x^*\|^2 - 2\bar{\gamma}\mathbf{E}\langle \bar{v}_k, x_k - x^* \rangle + 5\bar{\gamma}^2\mathbf{E}\|\nabla f(x_k)\|^2 + \mathbf{E}\|x_k - z_k\|^2 + \bar{\gamma}^2(5T_1 + 3T_2).$$

Next, since

$$\begin{aligned} T_1 &\stackrel{(6)}{\leq} \frac{1}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E}\|\nabla f^i(x_{k,l}^i) - \nabla f^i(x_k)\|^2 \\ &\stackrel{\text{Assumption 2}}{\leq} \frac{L^2}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E}\|x_{k,l}^i - x_k\|^2, \text{ and} \\ T_2 &\stackrel{(6)}{\leq} \frac{1}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E}\|g^i(x_{k,l}^i) - \nabla f^i(x_{k,l}^i)\|^2 \\ &\stackrel{\text{Assumption 3}}{\leq} \sigma^2, \end{aligned} \quad (12)$$

we have

$$\begin{aligned} \mathbf{E}\|z_{k+1} - x^*\|^2 &\leq \mathbf{E}\|z_k - x^*\|^2 - 2\bar{\gamma}\mathbf{E}\langle \bar{v}_k, x_k - x^* \rangle \\ &\quad + 5\bar{\gamma}^2\mathbf{E}\|\nabla f(x_k)\|^2 + \mathbf{E}\|x_k - z_k\|^2 \\ &\quad + \frac{5L^2\bar{\gamma}^2}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E}\|x_{k,l}^i - x_k\|^2 + 3\bar{\gamma}^2\sigma^2. \end{aligned} \quad (14)$$

Step 2) Bound $\mathbf{E}\langle \bar{v}_k, x_k - x^* \rangle$: Next, by the convexity of $f^i(x)$ and by L -Lipschitz continuity of $\nabla f^i(x)$,

$$\begin{aligned} -\langle \nabla f^i(x_{k,l}^i), x_{k,l}^i - x^* \rangle &\leq -[f^i(x_{k,l}^i) - f^i(x^*)], \text{ and} \\ -\langle \nabla f^i(x_{k,l}^i), x_k - x_{k,l}^i \rangle &\leq [f^i(x_{k,l}^i) - f^i(x_k)] + \frac{L}{2}\|x_k - x_{k,l}^i\|^2. \end{aligned}$$

By the above inequalities,

$$\begin{aligned} -\mathbf{E}\langle \bar{v}_k, x_k - x^* \rangle &= -\frac{1}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E}\langle \nabla f^i(x_{k,l}^i), x_k - x^* \rangle \\ &\leq -\mathbf{E}[f(x_k) - f(x^*)] + \frac{L}{2nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E}\|x_k - x_{k,l}^i\|^2. \end{aligned}$$

Plugging this result into (14) yields

$$\begin{aligned} V_{k+1} &\leq V_k - 2\bar{\gamma}W_k + 5\bar{\gamma}^2\mathbf{E}\|\nabla f(x_k)\|^2 + \mathbf{E}\|x_k - z_k\|^2 + 3\bar{\gamma}^2\sigma^2 \\ &\quad + \frac{5L^2\bar{\gamma}^2 + (L/2)\bar{\gamma}}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E}\|x_{k,l}^i - x_k\|^2, \end{aligned}$$

where $V_k := \mathbf{E}\|z_k - x^*\|^2$ and $W_k := \mathbf{E}[f(x_k) - f(x^*)]$. Since each $f^i(x)$ is convex and has L -Lipschitz continuous gradient, the function $f(x) = (1/n)\sum_{i=1}^n f^i(x)$ is convex and has L -Lipschitz continuous gradient, i.e. $\|\nabla f(x)\|^2 \leq 2L[f(x) - f(x^*)]$ for $x \in \mathbb{R}^d$. Therefore,

$$\begin{aligned} V_{k+1} &\leq V_k - 2\bar{\gamma}(1 - 5L\bar{\gamma})W_k + \mathbf{E}\|x_k - z_k\|^2 + 3\bar{\gamma}^2\sigma^2 \\ &\quad + \frac{5L^2\bar{\gamma}^2 + (L/2)\bar{\gamma}}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E}\|x_{k,l}^i - x_k\|^2. \end{aligned}$$

By the fact that $z_k = x_k + (1/n)\sum_{i=1}^n e_k^i$ and by (6),

$$\begin{aligned} V_{k+1} &\leq V_k - 2\bar{\gamma}(1 - 5L\bar{\gamma})W_k + \frac{1}{n} \sum_{i=1}^n \mathbf{E}\|e_k^i\|^2 + 3\bar{\gamma}^2\sigma^2 \\ &\quad + \frac{L\bar{\gamma}}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E}\|x_{k,l}^i - x_k\|^2. \end{aligned} \quad (15)$$

Step 3) Define a Lyapunov function, and use Lemma 2 and 3: We next define the Lyapunov function $\Theta_k = \mathbf{E}\|z_k - x^*\|^2 + (A/n)\sum_{i=1}^n \mathbf{E}\|e_k^i\|^2$ with $A > 0$. Then, by (15)

$$\begin{aligned} \Theta_{k+1} &\leq V_k - 2\bar{\gamma}(1 - 5L\bar{\gamma})W_k \\ &\quad + \frac{A}{n} \sum_{i=1}^n \mathbf{E}\|e_{k+1}^i\|^2 + \frac{1}{n} \sum_{i=1}^n \mathbf{E}\|e_k^i\|^2 + 3\bar{\gamma}^2\sigma^2 \\ &\quad + \frac{L\bar{\gamma}}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E}\|x_{k,l}^i - x_k\|^2. \end{aligned}$$

Next, by (11),

$$\begin{aligned} \Theta_{k+1} &\leq V_k - 2\bar{\gamma}(1 - 5L\bar{\gamma})W_k \\ &\quad + \frac{A(1 - \alpha/2) + 1}{n} \sum_{i=1}^n \mathbf{E}\|e_k^i\|^2 + 3\bar{\gamma}^2\sigma^2 \\ &\quad + \frac{6A(1 + 2/\alpha)(1 - \alpha)\bar{\gamma}^2}{n} \sum_{i=1}^n (\mathbf{E}\|\nabla f^i(x_k)\|^2 + \sigma^2) \\ &\quad + \frac{L\bar{\gamma}}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E}\|x_{k,l}^i - x_k\|^2. \end{aligned}$$

If $A = 2/\alpha$, then

$$\begin{aligned} \Theta_{k+1} &\leq \Theta_k - 2\bar{\gamma}(1 - 5L\bar{\gamma})W_k + 3\bar{\gamma}^2\sigma^2 \\ &\quad + \frac{6A(1 + 2/\alpha)(1 - \alpha)\bar{\gamma}^2}{n} \sum_{i=1}^n (\mathbf{E}\|\nabla f^i(x_k)\|^2 + \sigma^2) \\ &\quad + \frac{L\bar{\gamma}}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E}\|x_{k,l}^i - x_k\|^2. \end{aligned}$$

By (10) from Lemma 2, and by re-arranging the terms,

$$\begin{aligned} \Theta_{k+1} &\leq \Theta_k - 2\bar{\gamma}(1 - 5L\bar{\gamma})W_k \\ &\quad + \bar{\gamma}^2 \frac{\Gamma + 6L\bar{\gamma}}{n} \sum_{i=1}^n \mathbf{E}\|\nabla f^i(x_k)\|^2 \\ &\quad + \bar{\gamma}^2(3 + \Gamma + 6L\bar{\gamma})\sigma^2, \end{aligned}$$

where $\Gamma = (12/\alpha)(1 + 2/\alpha)(1 - \alpha)$. If $\gamma \leq 10L$, then

$$\begin{aligned} \Theta_{k+1} &\leq \Theta_k - \bar{\gamma}W_k + \bar{\gamma}^2 \frac{\Gamma + 1}{n} \sum_{i=1}^n \mathbf{E}\|\nabla f^i(x_k)\|^2 \\ &\quad + \bar{\gamma}^2(4 + \Gamma)\sigma^2. \end{aligned} \quad (16)$$

Step 4) Bound $(1/n) \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x_k)\|^2$: Next, by the fact that $\|x + y\|^2 \leq 2\|x\|^2 + 2\|y\|^2$ with $x = \nabla f^i(x_k) - \nabla f^i(x^*)$ and $y = \nabla f^i(x^*)$,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x_k)\|^2 &\leq \frac{2}{n} \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x_k) - \nabla f^i(x^*)\|^2 \\ &\quad + \frac{2}{n} \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x^*)\|^2. \end{aligned}$$

By the convexity and L -smoothness of $f^i(x)$, and by the fact that $f(x) = (1/n) \sum_{i=1}^n f_i(x)$, and that $\nabla f(x^*) = 0$,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \|\nabla f^i(x_k) - \nabla f^i(x^*)\|^2 &\leq \frac{2L}{n} \sum_{i=1}^n [f^i(x_k) - f^i(x^*) - \langle \nabla f^i(x^*), x_k - x^* \rangle] \\ &= 2L[f(x_k) - f(x^*)]. \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x_k)\|^2 &\leq 4L\mathbf{E}[f(x_k) - f(x^*)] + \frac{2}{n} \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x^*)\|^2, \end{aligned}$$

By plugging this result into (16),

$$\Theta_{k+1} \leq \Theta_k - \bar{\gamma}(1 - 4\beta_1 L \bar{\gamma})W_k + \bar{\gamma}^2 e,$$

where $e = \beta_2 \sigma^2 + \frac{2\beta_1}{n} \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x^*)\|^2$, $\beta_1 = \Gamma + 1$, and $\beta_2 = \Gamma + 4$. If $\bar{\gamma} \leq 1/(8\beta_1 L)$, then

$$\Theta_{k+1} \leq \Theta_k - \frac{\bar{\gamma}}{2} W_k + \bar{\gamma}^2 e.$$

Step 5) Conclude the convergence bound: By the convexity of $f(x)$,

$$\begin{aligned} \mathbf{E} \left[f \left(\frac{1}{K} \sum_{k=0}^{K-1} x_k \right) - f(x^*) \right] &\leq \frac{1}{K} \sum_{k=0}^{K-1} W_k \\ &\leq \frac{2}{\bar{\gamma}K} \sum_{k=0}^{K-1} [\Theta_k - \Theta_{k+1}] + 2\bar{\gamma}e \\ &\leq \frac{2\Theta_0}{\bar{\gamma}K} + 2\bar{\gamma}e. \end{aligned}$$

Finally, if $e_0^i = 0$ for all i , then $\Theta_0 = \|x_0 - x^*\|^2$.

APPENDIX C PROOF OF THEOREM 2

We derive the main result in the following steps.

Step 1) Use the Lipschitz continuous gradient of $f(x)$:

Since $\nabla f_i(x)$ is L -Lipschitz continuous, $\nabla f(x)$ is L -Lipschitz continuous. By the L -Lipschitz continuity of $\nabla f(x)$, and (4)

$$f(z_{k+1}) - f_{\inf} \leq f(z_k) - f_{\inf} - \bar{\gamma} \langle \nabla f(z_k), v_k \rangle + \frac{\bar{\gamma}^2 L}{2} \|v_k\|^2,$$

where $f_{\inf} = \inf_{x \in \mathbb{R}^d} f(x)$. Next, by taking the expectation, and by using the fact that $\mathbf{E}[v_k] = \bar{v}_k := (1/[nT]) \sum_{i=1}^n \sum_{l=0}^{T-1} \nabla f^i(x_{k,l}^i)$,

$$\begin{aligned} \mathbf{E}[f(z_{k+1}) - f_{\inf}] &\leq \mathbf{E}[f(z_k) - f_{\inf}] - \bar{\gamma} \mathbf{E} \langle \nabla f(z_k), \nabla f(x_k) \rangle \\ &\quad + \bar{\gamma} \mathbf{E} \langle \nabla f(z_k), \nabla f(x_k) - \bar{v}_k \rangle + \frac{\bar{\gamma}^2 L}{2} \mathbf{E} \|v_k\|^2. \end{aligned}$$

Since

$$\begin{aligned} \langle \nabla f(z_k), \nabla f(x_k) \rangle &\stackrel{(8)}{=} \frac{1}{2} \|\nabla f(z_k)\|^2 + \frac{1}{2} \|\nabla f(x_k)\|^2 \\ &\quad - \frac{1}{2} \|\nabla f(z_k) - \nabla f(x_k)\|^2, \quad \text{and} \\ \langle \nabla f(z_k), \nabla f(x_k) - \bar{v}_k \rangle &\stackrel{(7)}{\leq} \frac{\theta}{2} \|\nabla f(z_k)\|^2 + \frac{1}{2\theta} \|\nabla f(x_k) - \bar{v}_k\|^2, \end{aligned}$$

for $\theta > 0$, we get

$$\begin{aligned} V_{k+1} &\leq V_k - \frac{\bar{\gamma}}{2} \mathbf{E} \|\nabla f(z_k)\|^2 - \frac{\bar{\gamma}(1-\theta)}{2} \mathbf{E} \|\nabla f(x_k)\|^2 \\ &\quad + \frac{\bar{\gamma}}{2} \mathbf{E} \|\nabla f(z_k) - \nabla f(x_k)\|^2 + \frac{\bar{\gamma}}{2\theta} \mathbf{E} \|\nabla f(x_k) - \bar{v}_k\|^2 \\ &\quad + \frac{\bar{\gamma}^2 L}{2} \mathbf{E} \|v_k\|^2, \end{aligned}$$

where $V_k = \mathbf{E}[f(z_k) - f_{\inf}]$. Next, since

$$\begin{aligned} \mathbf{E} \|v_k\|^2 &\stackrel{(6)}{\leq} 3\mathbf{E} \|\nabla f(x_k)\|^2 + 3\mathbf{E} \|\bar{v}_k - v_k\|^2 \\ &\quad + 3\mathbf{E} \|\nabla f(x_k) - \bar{v}_k\|^2, \quad \text{and} \\ \|\nabla f(z_k) - \nabla f(x_k)\|^2 &\leq L^2 \|z_k - x_k\|^2, \end{aligned}$$

we have

$$\begin{aligned} V_{k+1} &\leq V_k - \frac{\bar{\gamma}}{2} \mathbf{E} \|\nabla f(z_k)\|^2 - \frac{\bar{\gamma}(1-\theta) - 3\bar{\gamma}^2 L}{2} \mathbf{E} \|\nabla f(x_k)\|^2 \\ &\quad + \frac{L^2 \bar{\gamma}}{2} \|z_k - x_k\|^2 + \left(\frac{\bar{\gamma}}{2\theta} + \frac{3\bar{\gamma}^2 L}{2} \right) \mathbf{E} \|\nabla f(x_k) - \bar{v}_k\|^2 \\ &\quad + \frac{3\bar{\gamma}^2 L}{2} \mathbf{E} \|\bar{v}_k - v_k\|^2. \end{aligned}$$

If $\theta = 1/2$, then

$$\begin{aligned} V_{k+1} &\leq V_k - \frac{\bar{\gamma}}{2} \mathbf{E} \|\nabla f(z_k)\|^2 - \frac{\bar{\gamma}/2 - 3\bar{\gamma}^2 L}{2} \mathbf{E} \|\nabla f(x_k)\|^2 \\ &\quad + \frac{L^2 \bar{\gamma}}{2} \|z_k - x_k\|^2 + \left(\bar{\gamma} + \frac{3\bar{\gamma}^2 L}{2} \right) \mathbf{E} \|\nabla f(x_k) - \bar{v}_k\|^2 \\ &\quad + \frac{3\bar{\gamma}^2 L}{2} \mathbf{E} \|\bar{v}_k - v_k\|^2. \end{aligned}$$

Next, by the upper-bound for $\mathbf{E} \|\bar{v}_k - \nabla f(x_k)\|^2$ in (12), and for $\mathbf{E} \|v_k - \bar{v}_k\|^2$ in (13),

$$\begin{aligned} V_{k+1} &\leq V_k - \frac{\bar{\gamma}}{2} \mathbf{E} \|\nabla f(z_k)\|^2 - \frac{\bar{\gamma}/2 - 3\bar{\gamma}^2 L}{2} \mathbf{E} \|\nabla f(x_k)\|^2 \\ &\quad + \frac{L^2 \bar{\gamma}}{2} \|z_k - x_k\|^2 + \frac{3\bar{\gamma}^2 L}{2} \sigma^2 \\ &\quad + \left(1 + \frac{3\bar{\gamma} L}{2} \right) \frac{\bar{\gamma} L^2}{nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E} \|x_{k,l}^i - x_k\|^2. \end{aligned}$$

If $\bar{\gamma} \leq 1/(12L)$, then

$$\begin{aligned} V_{k+1} \leq & V_k - \frac{\bar{\gamma}}{2} \mathbf{E} \|\nabla f(z_k)\|^2 - \frac{\bar{\gamma}}{8} \mathbf{E} \|\nabla f(x_k)\|^2 \\ & + \frac{L^2 \bar{\gamma}}{2} \|z_k - x_k\|^2 + \frac{3\bar{\gamma}^2 L}{2} \sigma^2 \\ & + \frac{9\bar{\gamma} L^2}{8nT} \sum_{i=1}^n \sum_{l=0}^{T-1} \mathbf{E} \|x_{k,l}^i - x_k\|^2. \end{aligned}$$

Step 2) Bound $\frac{1}{T} \sum_{l=0}^{T-1} \mathbf{E} \|x_{k,l}^i - x_k\|^2$: Next, by (10) from Lemma 2,

$$\begin{aligned} V_{k+1} \leq & V_k - \frac{\bar{\gamma}}{2} \mathbf{E} \|\nabla f(z_k)\|^2 - \frac{\bar{\gamma}}{8} \mathbf{E} \|\nabla f(x_k)\|^2 \\ & + \frac{L^2 \bar{\gamma}}{2} \|z_k - x_k\|^2 + \left(1 + \frac{9}{2} \bar{\gamma} L\right) \frac{3\bar{\gamma}^2 L}{2} \sigma^2 \\ & + \frac{27\bar{\gamma}^3 L^2}{4n} \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x_k)\|^2. \end{aligned}$$

Since $\bar{\gamma} \leq 1/(12L)$, we get

$$\begin{aligned} V_{k+1} \leq & V_k - \frac{\bar{\gamma}}{2} \mathbf{E} \|\nabla f(z_k)\|^2 - \frac{\bar{\gamma}}{8} \mathbf{E} \|\nabla f(x_k)\|^2 \\ & + \frac{L^2 \bar{\gamma}}{2} \|z_k - x_k\|^2 + \frac{33\bar{\gamma}^2 L}{16} \sigma^2 \\ & + \frac{27\bar{\gamma}^3 L^2}{4n} \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x_k)\|^2. \end{aligned} \quad (17)$$

Step 3) Bound $(1/n) \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x_k)\|^2$: Next, we bound $(1/n) \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x_k)\|^2$ by using (6) and Assumption 4 (data heterogeneity):

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x_k)\|^2 & \leq \frac{2}{n} \sum_{i=1}^n \mathbf{E} \|\nabla f(x_k)\|^2 \\ & \quad + \frac{2}{n} \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x_k) - \nabla f(x_k)\|^2 \\ & \leq 2\mathbf{E} \|\nabla f(x_k)\|^2 + 2\xi^2. \end{aligned} \quad (18)$$

Thus, plugging (18) into (17) yields

$$\begin{aligned} V_{k+1} \leq & V_k - \frac{\bar{\gamma}}{2} \mathbf{E} \|\nabla f(z_k)\|^2 - \left(\frac{\bar{\gamma}}{8} - \frac{27\bar{\gamma}^3 L^2}{2}\right) \mathbf{E} \|\nabla f(x_k)\|^2 \\ & + \frac{L^2 \bar{\gamma}}{2} \|z_k - x_k\|^2 + \frac{33\bar{\gamma}^2 L}{16} \sigma^2 + \frac{27\bar{\gamma}^3 L^2}{2} \xi^2. \end{aligned}$$

Since $\bar{\gamma} \leq 1/(12L)$, we have

$$\begin{aligned} V_{k+1} \leq & V_k - \frac{\bar{\gamma}}{2} \mathbf{E} \|\nabla f(z_k)\|^2 - \frac{\bar{\gamma}}{32} \mathbf{E} \|\nabla f(x_k)\|^2 \\ & + \frac{L^2 \bar{\gamma}}{2} \|z_k - x_k\|^2 + \bar{\gamma}^2 L e, \end{aligned}$$

where $e = (33/16)\sigma^2 + (9/8)\xi^2$. Next, by the fact that $z_k = x_k + (1/n) \sum_{i=1}^n e_k^i$, by (6), and by the non-negativity of $\|x\|^2$,

$$V_{k+1} \leq V_k - \frac{\bar{\gamma}}{32} \mathbf{E} \|\nabla f(x_k)\|^2 + \frac{L^2 \bar{\gamma}}{2n} \sum_{i=1}^n \mathbf{E} \|e_k^i\|^2 + \bar{\gamma}^2 L e. \quad (19)$$

Step 4) Define a Lyapunov function, and use Lemma 3: Next, define $\Theta_k = V_k + A \frac{1}{n} \sum_{i=1}^n \|e_k^i\|^2$ with $A > 0$. Then,

$$\begin{aligned} \Theta_{k+1} & \stackrel{(19)}{\leq} V_k - \frac{\bar{\gamma}}{32} \mathbf{E} \|\nabla f(x_k)\|^2 + \bar{\gamma}^2 L e \\ & \quad + \frac{A}{n} \sum_{i=1}^n \mathbf{E} \|e_{k+1}^i\|^2 + \frac{L^2 \bar{\gamma}}{2n} \sum_{i=1}^n \mathbf{E} \|e_k^i\|^2. \end{aligned}$$

Therefore,

$$\begin{aligned} \Theta_{k+1} & \stackrel{(11)}{\leq} V_k - \frac{\bar{\gamma}}{32} \mathbf{E} \|\nabla f(x_k)\|^2 + \bar{\gamma}^2 (L e + A \Gamma \sigma^2) \\ & \quad + \frac{A \Gamma \bar{\gamma}^2}{n} \sum_{i=1}^n \mathbf{E} \|\nabla f^i(x_k)\|^2 \\ & \quad + (A(1 - \alpha/2) + L^2 \bar{\gamma}/2) \frac{1}{n} \sum_{i=1}^n \mathbf{E} \|e_k^i\|^2, \end{aligned}$$

where $\Gamma = 6(1 + 2/\alpha)(1 - \alpha)$. Next, by (18),

$$\begin{aligned} \Theta_{k+1} \leq & V_k - \left(\frac{\bar{\gamma}}{32} - 2A\Gamma\bar{\gamma}^2\right) \mathbf{E} \|\nabla f(x_k)\|^2 \\ & + \bar{\gamma}^2 (L e + A\Gamma\sigma^2 + 2A\Gamma\xi^2) \\ & + (A(1 - \alpha/2) + L^2 \bar{\gamma}/2) \frac{1}{n} \sum_{i=1}^n \mathbf{E} \|e_k^i\|^2. \end{aligned}$$

If $A = L^2 \bar{\gamma}/\alpha$, then

$$\begin{aligned} \Theta_{k+1} \leq & \Theta_k - \left(\frac{\bar{\gamma}}{32} - \frac{2L^2 \Gamma \bar{\gamma}^3}{\alpha}\right) \mathbf{E} \|\nabla f(x_k)\|^2 \\ & + \bar{\gamma}^2 \bar{e}. \end{aligned}$$

where $\bar{e} = L e + (L\Gamma/[12\alpha])\sigma^2 + (L\Gamma/[6\alpha])\xi^2$. Since $\bar{\gamma} \leq 1/(12L)$, we have

$$\Theta_{k+1} \leq \Theta_k - \left(\frac{\bar{\gamma}}{32} - \frac{2L^2 \Gamma \bar{\gamma}^3}{\alpha}\right) \mathbf{E} \|\nabla f(x_k)\|^2 + \bar{\gamma}^2 \bar{e}.$$

If $\gamma \leq \frac{1}{8\sqrt{2}} \frac{\sqrt{\alpha}}{\sqrt{\Gamma}} \frac{1}{L}$, then

$$\Theta_{k+1} \leq \Theta_k - \frac{\bar{\gamma}}{64} \mathbf{E} \|\nabla f(x_k)\|^2 + \bar{\gamma}^2 \bar{e}. \quad (20)$$

Step 5) Conclude the convergence bound: Let $\hat{x}_K$ be sampled uniformly at random from the set $\{x_0, x_1, \dots, x_{K-1}\}$. Then,

$$\begin{aligned} \mathbf{E} \|\nabla f(\hat{x}_K)\|^2 & = \frac{1}{K} \sum_{l=0}^{K-1} \mathbf{E} \|\nabla f(x_l)\|^2 \\ & \stackrel{(20)}{\leq} \frac{64}{\bar{\gamma} K} \sum_{l=0}^{K-1} [\Theta_k - \Theta_{k+1}] + 64\bar{\gamma} \bar{e} \leq \frac{64\Theta_0}{\bar{\gamma} K} + 64\bar{\gamma} \bar{e}. \end{aligned}$$

Finally, if $e_0^i = 0$ for all i , then $\Theta_0 = f(x_0) - f_{\inf}$.

REFERENCES

- [1] N. Kimura and S. Latifi, "A survey on data compression in wireless sensor networks," in *International Conference on Information Technology: Coding and Computing (ITCC'05)-Volume II*, vol. 2. IEEE, 2005, pp. 8–13.
- [2] S. Khirirat, S. Magnússon, A. Aytikin, and M. Johansson, "A flexible framework for communication-efficient machine learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, 2021, pp. 8101–8109.

- [3] F. Seide, H. Fu, J. Droppo, G. Li, and D. Yu, "1-bit stochastic gradient descent and its application to data-parallel distributed training of speech dnn's," in *Fifteenth annual conference of the international speech communication association*, 2014.
- [4] H. Tang, S. Gan, C. Zhang, T. Zhang, and J. Liu, "Communication compression for decentralized training," *Advances in Neural Information Processing Systems*, vol. 31, 2018.
- [5] S. U. Stich, J.-B. Cordonnier, and M. Jaggi, "Sparsified sgd with memory," *Advances in Neural Information Processing Systems*, vol. 31, 2018.
- [6] P. Richtárik, I. Sokolov, and I. Fatkhullin, "Ef21: A new, simpler, theoretically better, and practically faster error feedback," *Advances in Neural Information Processing Systems*, vol. 34, pp. 4384–4396, 2021.
- [7] T. Sun, D. Li, and B. Wang, "Decentralized federated averaging," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022.
- [8] S. Li, Q. Qi, J. Wang, H. Sun, Y. Li, and F. R. Yu, "Ggs: General gradient sparsification for federated learning in edge computing," in *ICC 2020-2020 IEEE International Conference on Communications (ICC)*. IEEE, 2020, pp. 1–7.
- [9] H. Sun, X. Ma, and R. Q. Hu, "Adaptive federated learning with gradient compression in uplink noma," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 12, pp. 16 325–16 329, 2020.
- [10] S. Vaishnav, M. Efthymiou, and S. Magnússon, "Energy-efficient and adaptive gradient sparsification for federated learning," in *ICC 2023 - IEEE International Conference on Communications*, 2023, pp. 1256–1261.
- [11] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Artificial intelligence and statistics*. PMLR, 2017, pp. 1273–1282.
- [12] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," *Proceedings of Machine learning and systems*, vol. 2, pp. 429–450, 2020.
- [13] R. Pathak and M. J. Wainwright, "Fedsplit: An algorithmic framework for fast federated optimization," *Advances in neural information processing systems*, vol. 33, pp. 7057–7066, 2020.
- [14] X. Zhang, M. Hong, S. Dhople, W. Yin, and Y. Liu, "Fedpd: A federated learning framework with adaptivity to non-iid data," *IEEE Transactions on Signal Processing*, vol. 69, pp. 6055–6070, 2021.
- [15] K. Mishchenko, G. Malinovsky, S. Stich, and P. Richtárik, "Proxskip: Yes! local gradient steps provably lead to communication acceleration! finally!" in *International Conference on Machine Learning*. PMLR, 2022, pp. 15 750–15 769.
- [16] A. Reiszadeh, A. Mokhtari, H. Hassani, A. Jadbabaie, and R. Pedarsani, "Fedpaq: A communication-efficient federated learning method with periodic averaging and quantization," in *International Conference on Artificial Intelligence and Statistics*. PMLR, 2020, pp. 2021–2031.
- [17] D. Basu, D. Data, C. Karakus, and S. Diggavi, "Qsparse-local-SGD: distributed SGD with quantization, sparsification and local computations," *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [18] F. Haddadpour, M. M. Kamani, A. Mokhtari, and M. Mahdavi, "Federated learning with compression: Unified analysis and sharp guarantees," in *International Conference on Artificial Intelligence and Statistics*. PMLR, 2021, pp. 2350–2358.
- [19] D. Alistarh, T. Hoeffler, M. Johansson, N. Konstantinov, S. Khirirat, and C. Renggli, "The convergence of sparsified gradient methods," *Advances in Neural Information Processing Systems*, vol. 31, 2018.
- [20] F. Sattler, S. Wiedemann, K.-R. Müller, and W. Samek, "Robust and communication-efficient federated learning from non-iid data," *IEEE transactions on neural networks and learning systems*, vol. 31, no. 9, pp. 3400–3413, 2019.
- [21] J. Konečný, H. B. McMahan, F. X. Yu, P. Richtárik, A. T. Suresh, and D. Bacon, "Federated learning: Strategies for improving communication efficiency," *arXiv preprint arXiv:1610.05492*, 2016.
- [22] D. Alistarh, D. Grubic, J. Li, R. Tomioka, and M. Vojnovic, "Qsgd: Communication-efficient sgd via gradient quantization and encoding," *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [23] J. Bernstein, Y.-X. Wang, K. Azizzadenesheli, and A. Anandkumar, "signsgd: Compressed optimisation for non-convex problems," in *International Conference on Machine Learning*. PMLR, 2018, pp. 560–569.
- [24] A. T. Suresh, X. Y. Felix, S. Kumar, and H. B. McMahan, "Distributed mean estimation with limited communication," in *International Conference on Machine Learning*. PMLR, 2017, pp. 3329–3337.
- [25] W. Wen, C. Xu, F. Yan, C. Wu, Y. Wang, Y. Chen, and H. Li, "Terngrad: Ternary gradients to reduce communication in distributed deep learning," *Advances in neural information processing systems*, vol. 30, 2017.
- [26] A. Mora, L. Foschini, and P. Bellavista, "Structured sparse ternary compression for convolutional layers in federated learning," in *2022 IEEE 95th Vehicular Technology Conference: (VTC2022-Spring)*, 2022, pp. 1–5.
- [27] A. Kozłowski and J. Sosnowski, "Analysing efficiency of ipv6 packet transmission over 6lowpan network," in *Photonics Applications in Astronomy, Communications, Industry, and High Energy Physics Experiments 2017*, vol. 10445. SPIE, 2017, pp. 456–467.
- [28] C. E. Shannon, "Communication in the presence of noise," *Proceedings of the IRE*, vol. 37, no. 1, pp. 10–21, 1949.
- [29] I. Sarkar, M. Adhikari, N. Kumar, and S. Kumar, "Dynamic task placement for deadline-aware iot applications in federated fog networks," *IEEE Internet of Things Journal*, vol. 9, no. 2, pp. 1469–1478, 2022.
- [30] Y. LeCun, "The mnist database of handwritten digits," <http://yann.lecun.com/exdb/mnist/>, 1998.
- [31] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms," *arXiv preprint arXiv:1708.07747*, 2017.
- [32] S. Wang, T. Tuor, T. Salonidis, K. K. Leung, C. Makaya, T. He, and K. Chan, "Adaptive federated learning in resource constrained edge computing systems," *IEEE journal on selected areas in communications*, vol. 37, no. 6, pp. 1205–1221, 2019.
- [33] S. Khirirat, M. Johansson, and D. Alistarh, "Gradient compression for communication-limited convex optimization," in *IEEE Conference on Decision and Control (CDC)*. IEEE, 2018, pp. 166–171.

Shubham Vaishnav is currently pursuing Ph.D. at the Department of Computer and Systems Science, Stockholm University, Sweden. This research program is funded by Digital Futures. He received Bachelors' and Masters' degrees in Computer Science and Engineering from the Indian Institute of Technology (Indian School of Mines), Dhanbad. His research interests include Multiobjective and Adaptive decision-making in IoT and Edge networks, Reinforcement learning, Federated Learning, and data-driven decision-making in critical societal infrastructure.

Sarit Khirirat received his B.Eng. in Electrical Engineering from Chulalongkorn University, Thailand, in 2013; the Master's degree in Systems, Control and Robotics from KTH Royal Institute of Technology, Sweden, in 2017; and the PhD at the Division of Decision and Control Systems from the same institution in 2022, supported by the Wallenberg AI, Autonomous Systems and Software program, Sweden's largest individual research funding program. He is currently a postdoctoral fellow at King Abdullah University of Science and Technology (KAUST). His research interests include distributed optimization algorithms for federated learning applications.

Sindri Magnússon is an Associate Professor in the Department of Computer and Systems Science at Stockholm University, Sweden. He received a B.Sc. degree in Mathematics from the University of Iceland, Reykjavik Iceland, in 2011, a Masters degree in Applied Mathematics (Optimization and Systems Theory) from KTH Royal Institute of Technology, Stockholm, Sweden, in 2013, and the PhD in Electrical Engineering from the same institution, in 2017. He was a postdoctoral researcher 2018–2019 at Harvard University, Cambridge, MA, and a visiting PhD student at Harvard University for 9 months in 2015 and 2016. His research interests include large-scale distributed/parallel optimization, machine learning, and control.